THE SHADOW OBSTRUCTION TOWER

RAEEZ LORGAT

ABSTRACT. The chiral bar construction of a vertex algebra $\mathcal{A}$ on an algebraic curve produces a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the modular convolution algebra, encoding the full genus expansion. The degree-r projections of $\Theta_{\mathcal{A}}$ form the *shadow obstruction tower*, an infinite sequence of invariants $S_2, S_3, S_4, \ldots$ determined by the operator product expansion.

We prove five results. First, the weighted generating function $H(t) = \sum_{r \geq 2} r\, S_r\, t^r$ satisfies $H(t)^2 = t^4\, Q_L(t)$, where the *shadow metric* $Q_L(t) = 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2$ is a quadratic polynomial: the tower is algebraic of degree 2, and every S_r for $r \geq 5$ is determined by three seeds (κ, α, S_4). Second, the critical discriminant $\Delta = 8\kappa S_4$ partitions all chirally Koszul algebras into four depth classes: Gaussian ($d_{\mathrm{alg}} = 0$), Lie ($d_{\mathrm{alg}} = 1$), contact ($d_{\mathrm{alg}} = 2$), and mixed ($d_{\mathrm{alg}} = \infty$). Third, the algebraic depth satisfies $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$; no value 3 or any finite value ≥ 3 is realized. Fourth, a chirally Koszul vertex algebra is Swiss-cheese formal if and only if it belongs to class $\mathbf{G}$. Fifth, the shadow obstruction tower is the Σ_n-coinvariant image of the ordered R-matrix tower in the E_1 convolution algebra.

We compute explicit shadow towers for all standard families (Heisenberg, affine Kac–Moody, $\beta\gamma$, Virasoro, $\mathcal{W}_N$), extract the quartic contact invariant $Q^{\mathrm{contact}}_{\mathrm{Vir}} = 10/[c(5c+22)]$, determine the shadow connection and its Koszul monodromy -1, and compute the genus-2 cross-channel correction $\delta F_2(\mathcal{W}_3) = (c+204)/(16c)$ by graph-by-graph stable graph summation over the seven genus-2 stable graphs.

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1. Introduction

1.1. The problem. Classical Koszul duality is a phenomenon of graded algebras over a field. A quadratic algebra A determines a dual coalgebra $A^!$; the bar construction $B(A) = T^c(s^{-1}\bar{A})$ mediates between them; and the Koszul property (bar cohomology concentrated in bar degree 1) is equivalent to the existence of a minimal resolution. The theory is the subject of Priddy [16], Beilinson–Ginzburg–Soergel [4], and Loday–Vallette [15].

On an algebraic curve X, the generators become sections of a $\mathcal{D}_X$-module, the quadratic relations become operator product expansions with meromorphic singularities, and the bar differential becomes an integral transform whose kernel is the logarithmic 1-form $\eta_{ij} = d\log(z_i - z_j)$ on the Fulton–MacPherson compactification $\mathrm{FM}_n(X)$. The resulting object, the *chiral bar complex* $\bar{B}^{\mathrm{ch}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, is a factorization coalgebra on the Ran space $\mathrm{Ran}(X)$ [3, 5, 8].

At genus 0, the classical theory embeds via restriction to the formal disk. The embedding is not an equivalence: the configuration-space geometry of $\mathrm{Conf}_n(\mathbb{A}^1)$, the Arnol'd algebra of logarithmic forms, the Fulton–MacPherson compactifications, and the E_1/E_∞ bar distinction are already present on $\mathbb{A}^1$ but absent over a bare point; and the passage from the formal disk to a point requires specifying a retraction and its homotopy transfer data. At genus $g \geq 1$, the topology of the curve forces curvature: the fiberwise bar differential satisfies $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g$, where $\omega_g = c_1(\mathbb{E})$ is the Hodge class on $\overline{\mathcal{M}}_g$ and $\kappa(\mathcal{A}) \in \mathbb{Q}(c)$ is an intrinsic invariant called the *modular characteristic*. This curvature has no analogue in the classical setting; its origin is the nontriviality of the Hodge bundle over $\overline{\mathcal{M}}_g$.

The present paper addresses a single question: *what invariants does the chiral bar construction produce beyond the modular characteristic, and what constrains them?*

1.2. The answer.

Three invariants suffice. A *primary line* is a one-parameter subfamily of the modular deformation space obtained by varying a single coupling constant while holding all others fixed: for Heisenberg, the level k; for Virasoro, the central charge c; for affine Kac–Moody, the level k at fixed Lie type $\mathfrak{g}$. Along each primary line, the chiral bar construction produces an infinite sequence of *shadow invariants* $S_2, S_3, S_4, \ldots$, and the entire sequence is algebraic of degree 2: a single quadratic polynomial

$$Q_L(t) \;=\; 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2,$$

the *shadow metric*, determines all S_r for $r \geq 5$ through the relation $H(t)^2 = t^4\, Q_L(t)$, where $H(t) = \sum_{r\geq 2} r\, S_r\, t^r$ is the weighted shadow generating function.

The critical discriminant $\Delta := 8\kappa S_4$ controls the depth of the tower: when $\Delta = 0$, the generating function is polynomial and the tower terminates; when $\Delta \neq 0$, the generating function has irrational Taylor coefficients and the tower extends to infinity. The transition is sharp: no intermediate depth is possible.

1.3. Main results.

Theorem 1.1 (Riccati algebraicity; Theorem 4.1). *Let $\mathcal{A}$ be a chirally Koszul vertex algebra, L a primary line (e.g. $\mathcal{H}_k$ parametrised by level k, or Vir_c parametrised by central charge c) with shadow data S_r, and $\kappa = S_2 \neq 0$. Then*

$$H(t)^2 \;=\; t^4\, Q_L(t).$$

Every S_r with $r \geq 5$ is determined by (κ, α, S_4) via $S_r = \frac{1}{r}[t^{r-2}]\sqrt{Q_L(t)}$.

Theorem 1.2 (Four-class partition; Theorem 5.1). *On any primary line L with $\kappa \neq 0$, the shadow depth $r_{\max} := \sup\{r : S_r \neq 0\}$ belongs to $\{2, 3, \infty\}$:*

*(i) **Class G** (Gaussian): $\Delta = 0$, $\alpha = 0$; $r_{\max} = 2$.*
*(ii) **Class L** (Lie): $\Delta = 0$, $\alpha \neq 0$; $r_{\max} = 3$.*
*(iii) **Class M** (mixed): $\Delta \neq 0$; $r_{\max} = \infty$.*

*A fourth class, **class C** (contact, $r_{\max} = 4$), arises when the single-line tower terminates but a quartic contact invariant from a two-field interaction survives. The archetype is the $\beta\gamma$ system (see Section 5.2 for the explicit computation).*

Theorem 1.3 (Depth gap; Proposition 6.2). *The algebraic depth satisfies $d_{\mathrm{alg}}(\mathcal{A}) \in \{0, 1, 2, \infty\}$. No value 3 or any finite value ≥ 3 is realized on the standard landscape. The four values correspond bijectively to the classes **G**, **L**, **C**, **M**.*

Theorem 1.4 (SC-formality characterizes class **G**; Proposition 7.1). *A chirally Koszul vertex algebra $\mathcal{A}$ in the standard landscape is Swiss-cheese formal ($m_k^{\mathrm{SC}} = 0$ for all $k \geq 3$) if and only if $\mathcal{A}$ belongs to class* **G**.

Theorem 1.5 (E_1 framing; Proposition 8.1). *The shadow obstruction tower $\{S_r\}_{r\geq 2}$ is the Σ_n-coinvariant image of the ordered R-matrix tower in the E_1 convolution algebra $\mathfrak{g}_{\mathcal{A}}^{E_1}$: at degree 2, $\kappa = \mathrm{av}(r(z))$ for abelian algebras, and $\kappa = \mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$ for non-abelian affine Kac–Moody (where the Sugawara shift contributes).*

The shadow obstruction tower also carries a logarithmic connection $\nabla^{\mathrm{sh}} = d - Q_L'/(2Q_L)\, dt$ with regular singularities at the zeros of Q_L; the monodromy around each branch point is -1 (the Koszul sign).

1.4. **Discussion of the results.** Each of the five main results captures a different facet of the shadow obstruction tower.

The algebraicity theorem (Theorem 1.1) is the structural core: it says that the MC recursion, which is an infinite system of quadratic equations (one at each degree), is equivalent to a single quadratic polynomial relation. The proof converts the recursion into a convolution identity by the substitution $a_n = (n+2)S_{n+2}$ and shows that the convolution $F(t)^2 = Q_L(t)$ holds at all orders. The key insight is that the MC recursion at degree r is precisely the statement that the degree-$(r-2)$ convolution vanishes: the recursion *is* the algebraicity.

The four-class partition (Theorem 1.2) is an immediate consequence: the factorization of the shadow metric Q_L controls the depth. The classification is exhaustive for the standard landscape and sharp: no intermediate depths are possible (Theorem 1.3).

The depth gap theorem (Theorem 1.3) has a clean algebraic explanation: on a single primary line, the shadow generating function is a degree-2 algebraic function, which is either rational (tower finite) or irrational (tower infinite). The contact class **C** arises by stratum separation: one additional degree from a transverse direction. No further finite extension is possible.

The SC-formality theorem (Theorem 1.4) connects the shadow tower to the operadic structure: the Swiss-cheese operations m_k^{SC} vanish for $k \geq 3$ if and only if the shadow tower terminates at degree 2. The proof uses the tree-shadow correspondence: the degree-r shadow is the closed-colour projection of the degree-r mixed SC tree.

The E_1 framing (Theorem 1.5) identifies the shadow tower as a projection of a richer structure: the ordered R-matrix tower in the E_1 convolution algebra. The averaging map av is lossy: it discards the spectral parameter, the R-matrix, and the Yangian. The shadow tower is the Σ_n-coinvariant residue.

For the $\mathcal{W}_3$ algebra at genus 2, the multi-channel stable graph sum produces a cross-channel correction $\delta F_2(\mathcal{W}_3) = (c+204)/(16c)$, resolving the multi-generator universality problem negatively: the scalar formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ fails at genus ≥ 2 for multi-weight algebras (UNIFORM-WEIGHT tag required; the correct decomposition is $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$). The computation is graph-by-graph: four of the seven genus-2 stable graphs contribute nonzero mixed-channel amplitudes, with the tadpole contribution $1/16$ being the unique c-independent term.

1.5. **The E_1 perspective.** The shadow obstruction tower is a projection. The primitive object is the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with its deconcatenation coproduct and spectral R-matrix $r(z) \in \mathfrak{g}_{\mathcal{A}}^{E_1}$. The shadow tower $\{S_r\}$ and the algebraicity relation $H(t)^2 = t^4\, Q_L(t)$ are statements about the Σ_n-coinvariant image: the degree-2 scalar κ, given by $\mathrm{av}(r(z))$ in abelian and scalar families and by $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$ for non-abelian affine Kac–Moody, and its higher-degree extensions under the lossy averaging map $\mathrm{av}\colon \mathfrak{g}_{\mathcal{A}}^{E_1} \to \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$.

The full E_1 tower retains the spectral parameter z, the R-matrix, and the Yangian structure; the shadow tower discards the spectral parameter by averaging. Everything in this paper is the Σ_n-coinvariant projection of an E_1 object. The ordered bar is the primitive; the shadow tower is the shadow.

1.6. **Context.** The algebraicity relation $H(t)^2 = t^4\, Q_L(t)$ is structurally parallel to the loop equations of matrix models [2, 6] and to the Eynard–Orantin topological recursion [7]: in those frameworks the spectral curve is an input; here the spectral curve $\Sigma = \{H^2 = t^4\, Q_L\}$ is an *output* of the algebraicity theorem, determined by the vertex algebra.

The shadow connection has Stokes data controlled by the critical discriminant: monodromy is trivial for classes **G** and **L** and nontrivial for class **M**, in parallel with the wall-crossing formalism of Kontsevich–Soibelman [14].

Classical Koszul duality over a field embeds into the genus-0, degree-2, $\Delta = 0$ stratum (class **G**, where the tower terminates immediately) via restriction to the formal disk. Everything beyond class **G** is new.

1.7. **Notation and conventions.** Throughout, X denotes a smooth projective algebraic curve over a field of characteristic zero. All differentials have cohomological degree +1. The bar construction uses desuspension (s^{-1}, degree -1).

A *vertex algebra* is a vertex algebra in the sense of [10, 13], equipped with a conformal vector, a conformal grading $\mathcal{A} = \bigoplus_{h\geq 0} \mathcal{A}_h$ with $\dim \mathcal{A}_h < \infty$, and the OPE $a(z)b(w) \sim \sum_{n\geq 0}(a_{(n)}b)(w)\cdot (z-w)^{-n-1}$.

The *modular characteristic* $\kappa(\mathcal{A}) \in \mathbb{Q}(c)$ is the genus-1 curvature scalar: $F_1(\mathcal{A}) = \kappa(\mathcal{A})/24$. The Hodge class $\lambda_g := c_1(\mathbb{E}_g) \in H^2(\overline{\mathcal{M}}_g, \mathbb{Q})$ is the first Chern class of the Hodge bundle. The Faber–Pandharipande coefficients are $\lambda_g^{\mathrm{FP}} = \int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2}\,\lambda_g$, with $\lambda_1^{\mathrm{FP}} = 1/24$ and $\lambda_2^{\mathrm{FP}} = 7/5760$.

1.8. **Guide to the reader.** The paper is organized around the shadow obstruction tower as the central object, constructed from first principles and then analyzed through five independent lenses.

Section 2 constructs the chiral bar complex and the Maurer–Cartan element $\Theta_{\mathcal{A}}$, beginning with the Heisenberg algebra as the motivating example. Section 3 defines the shadow obstruction tower as the sequence of degree projections of $\Theta_{\mathcal{A}}$ and establishes the master equation. Section 4 proves the Riccati algebraicity theorem: the conversion of the MC recursion into a convolution identity, the degree-by-degree verification, and the closed-form extraction of shadow coefficients. Section 5 derives the four-class partition from the factorization of the shadow metric. Section 6 proves the depth gap theorem, establishing that $d_{\mathrm{alg}} = 3$ is impossible. Section 7 proves that SC-formality characterizes class **G**. Section 8 develops the E_1 framing: the shadow tower as the averaged projection of the ordered R-matrix tower. Sections 9 and 10 extract the shadow connection and compute explicit towers for all standard families. Section 12 carries out the $\mathcal{W}_3$ cross-channel computation at genus 2.

The logical dependencies are: §2 $\to$ §3 $\to$ §4 $\to$ §5 $\to$ §6; the sections on SC-formality, E_1 framing, the shadow connection, and the genus-2 computation are logically independent of each other and may be read in any order after Section 5.

1.9. **Relation to companion papers.** The c-independent identity $S_3(\mathrm{Vir}_c) = 2$ is proved in [18]. The seven-face programme connecting the shadow tower to Gaudin models, Sklyanin brackets, and commuting Hamiltonians is developed in [19]. The present paper is self-contained: all results are proved from the chiral bar construction without reference to the companion papers.

2. The chiral bar complex and Maurer–Cartan element

2.1. **The chiral bar complex.** Let $\mathcal{A}$ be a vertex algebra on a smooth projective curve X. The Ran space $\mathrm{Ran}(X) = \coprod_{n\geq 1} X^n/\Sigma_n$ parameterizes nonempty finite subsets of X. For each $n \geq 1$, the Fulton–MacPherson compactification $\mathrm{FM}_n(X)$ resolves the diagonals in X^n by iterated blowups along all partial diagonals, producing a smooth manifold-with-corners whose boundary strata are indexed by trees of collisions [11].

Definition 2.1 (Chiral bar complex)**.** The *chiral bar complex* $B(\mathcal{A})$ is the factorization coalgebra on $\mathrm{Ran}(X)$ with:

(i) *Underlying space*: $B(\mathcal{A})_n := (s^{-1}\bar{\mathcal{A}})^{\otimes n}$, where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal and s^{-1} denotes desuspension (degree -1).

(ii) *Bar differential*: the coderivation d_B whose components are the collision residues

$$d_B|_{B(\mathcal{A})_n \to B(\mathcal{A})_{n-1}} \;=\; \sum_{i<j} \mathrm{Res}_{z_i\to z_j}\, a_i(z_i)\, a_j(z_j)\, d\log(z_i - z_j),$$

extracting the OPE residue along the logarithmic differential $d\log(z_i - z_j)$.

(iii) *Coproduct*: the cofactorization structure induced by partition of points into clusters.

The identity $d_B^2 = 0$ encodes OPE associativity.

Remark 2.2 (The $d\log$ absorption)*.* The bar differential extracts residues along $d\log(z - w)$, not along $dz/(z - w)$. The logarithmic differential absorbs one power of $(z - w)$: if the OPE has poles at $(z - w)^{-n-1}$, the collision residue has poles at $(z - w)^{-n}$. For Virasoro, the OPE has a quartic pole; the bar r-matrix has a cubic pole $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$.

Remark 2.3 (The ordered bar as primitive)*.* The bar complex of Definition 2.1 is the symmetric (factorization) bar $B^{\Sigma}(\mathcal{A})$, the Σ_n-coinvariant quotient of the ordered bar $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$. The ordered bar carries a deconcatenation coproduct (coassociative, not cocommutative) and supports the spectral R-matrix $r(z) \in \mathfrak{g}_{\mathcal{A}}^{E_1}$. The shadow obstruction tower $\{S_r\}$ and the algebraicity relation $H(t)^2 = t^4 Q_L(t)$ are statements about the Σ_n-coinvariant image: the degree-2 scalar κ, given by $\mathrm{av}(r(z))$ in abelian and scalar families and by $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$ for non-abelian affine Kac–Moody, and its higher-degree extensions. The ordered bar is the primitive object; the shadow tower is its averaged projection.

2.2. **Chirally Koszul algebras.**

Definition 2.4 (Chirally Koszul)**.** A vertex algebra $\mathcal{A}$ is *chirally Koszul* if the bar cohomology $H^\bullet(B(\mathcal{A}), d_B)$ is concentrated in bar degree 1: for every $n \geq 2$, the map $H^\bullet(B(\mathcal{A})_n) \to H^\bullet(B(\mathcal{A})_1)^{\otimes n}$ induced by the coproduct is a quasi-isomorphism.

The standard families (Heisenberg, affine Kac–Moody at non-critical level $k \neq -h^\vee$, $\beta\gamma$, Virasoro, and $\mathcal{W}_N$) are all chirally Koszul.

2.3. **The modular convolution algebra.**

Definition 2.5 (Modular cyclic deformation complex)**.** The *modular cyclic deformation complex* of $\mathcal{A}$ is

$$\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}) \;:=\; \prod_{\substack{g,n \\ 2g-2+n>0}} \mathrm{CoDer}^{\mathrm{cyc}}\big(B^{(g,n)}(\mathcal{A})\big)[1],$$

where the product ranges over all stable pairs. The differential is $d = d_{\mathrm{int}} + d_{\mathrm{sew}}$, with d_{int} the internal cyclic coderivation differential and d_{sew} the clutching operation from boundary maps on $\overline{\mathcal{M}}_{g,n}$. The Lie bracket is *graph composition*: for coderivations ξ and η,

$$[\xi, \eta] \ := \ \sum_{\Gamma} \xi \circ_{\Gamma} \eta, \tag{2.1}$$

summed over stable graphs connecting an output of ξ to an input of η.

2.4. The Maurer–Cartan element.

Proposition 2.6 (Bar-intrinsic MC element). *Let $D_{\mathcal{A}}$ denote the full bar differential on $B(\mathcal{A})$ and d_0 its genus-0 linear part. Define*

$$\Theta_{\mathcal{A}} \ := \ D_{\mathcal{A}} - d_0 \ \in \ \mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}).$$

Then $\Theta_{\mathcal{A}}$ satisfies the Maurer–Cartan equation

$$d_0\, \Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0.$$

Proof. The identity $D^2_{\mathcal{A}} = 0$ is a formal consequence of the boundary-of-boundary relation $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$. Substituting $D_{\mathcal{A}} = d_0 + \Theta_{\mathcal{A}}$ into $D^2_{\mathcal{A}} = 0$ and expanding:

$$0 \ = \ d_0^2 + d_0\, \Theta_{\mathcal{A}} + \Theta_{\mathcal{A}} \circ d_0 + \Theta_{\mathcal{A}} \circ \Theta_{\mathcal{A}}.$$

Since $d_0^2 = 0$ (OPE associativity) and the remaining terms identify with $d_0\, \Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}]$ in the cyclic coderivation Lie algebra, the result follows. □

3. The shadow obstruction tower

3.1. Definition and master equation.

Definition 3.1 (Shadow obstruction tower). The *shadow obstruction tower* of $\mathcal{A}$ is the sequence of truncations $\Theta^{\le r}_{\mathcal{A}}$ $(r \ge 2)$ obtained by retaining only the components of degree at most r in $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$. The *degree-r shadow* is the projection

$$\mathrm{Sh}_r(\mathcal{A}) \ := \ \pi_r(\Theta_{\mathcal{A}}) \ \in \ \mathrm{CoDer}^{\mathrm{cyc}}(B^{(\bullet,r)}(\mathcal{A}))[1].$$

Proposition 3.2 (All-degree master equation). *For each $r \ge 2$, the degree-r projection of the MC equation gives*

$$d_0(\mathrm{Sh}_r) + \sum_{\substack{j+k=r+2\\ j,k\ge 2}} [\mathrm{Sh}_j, \mathrm{Sh}_k] \ = \ 0.$$

The shadow Sh_r is determined recursively by $\mathrm{Sh}_2, \ldots, \mathrm{Sh}_{r-1}$ and a cohomological obstruction.

Proof. The differential d_0 and the bracket $[-,-]$ respect the degree filtration. Restricting the MC equation to degree r: the linear term is $d_0(\mathrm{Sh}_r)$, and the quadratic term at degree r receives contributions from pairs (j,k) with $j + k = r + 2$ (the bracket consumes two inputs via the internal edge, reducing total degree by 2). □

3.2. Primary lines and shadow data.

Definition 3.3 (Primary line and shadow data). A *primary line* in $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$ is a one-dimensional subspace $L = k \cdot \eta$ spanned by a single primary field η of definite conformal weight. On L, the degree-r shadow restricts to a scalar: $\mathrm{Sh}_r|_L = S_r \cdot x^r$, where x is the coordinate on L and $S_r = S_r(\mathcal{A}) \in \mathbb{Q}(c)$.

The first three scalars are:

- $S_2 =: \kappa$, the *modular characteristic*: $F_1(\mathcal{A}) = \kappa/24$.

- $S_3 =: \alpha$, the *cubic shadow*: the three-point collision residue on $\mathrm{FM}_3(X)$.
- S_4, the *quartic shadow*: the four-point collision residue, the first nonlinear OPE invariant.

3.3. **The shadow metric.**

Definition 3.4 (Shadow metric and critical discriminant). Let L be a primary line with $\kappa = S_2 \neq 0$ and $\alpha = S_3$. Define the *weighted initial data* $a_0 := 2\kappa$, $a_1 := 3\alpha$, $a_2 := 4S_4$. The *shadow metric* on L is

$$Q_L(t) \;:=\; a_0^2 + 2a_0a_1\,t + (a_1^2 + 2a_0a_2)\,t^2 \;=\; 4\kappa^2 + 12\kappa\alpha\,t + (9\alpha^2 + 16\kappa S_4)\,t^2.$$

The *critical discriminant* is

$$\Delta \;:=\; 8\kappa S_4.$$

Lemma 3.5 (Gaussian decomposition).

$$Q_L(t) \;=\; (2\kappa + 3\alpha\,t)^2 \;+\; 2\Delta\,t^2.$$

The first term is the shadow metric of the truncated tower with $S_r = 0$ for $r \geq 4$. The second term is the quartic interaction correction.

Proof. $(2\kappa + 3\alpha t)^2 = 4\kappa^2 + 12\kappa\alpha\,t + 9\alpha^2\,t^2$. Adding $2\Delta\,t^2 = 16\kappa S_4\,t^2$ recovers $Q_L(t)$. □

3.4. **The obstruction classes.** The MC equation at degree r takes the form $d_0(\mathrm{Sh}_r) = -\frac{1}{2}\sum_{j+k=r+2}[\mathrm{Sh}_j, \mathrm{Sh}_k]$. The right-hand side is a cocycle in the cyclic deformation complex (it is d_0-closed by the Jacobi identity for $[-,-]$ and the MC equation at lower degrees). The shadow coefficient S_r exists if and only if this cocycle is a coboundary.

Definition 3.6 (Obstruction classes). The *degree-r obstruction class* is

$$o_r(\mathcal{A}) \;:=\; \Big[-\frac{1}{2}\sum_{\substack{j+k=r+2\\ j,k\geq 2}}[\mathrm{Sh}_j, \mathrm{Sh}_k]\Big] \;\in\; H^2(\mathcal{A}^{\mathrm{sh}}_{r,0}),$$

the cohomology class of the quadratic obstruction in the shadow algebra $\mathcal{A}^{\mathrm{sh}} := H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}))$. When $o_r = 0$, the shadow coefficient S_r exists and is determined (up to a coboundary, which is fixed by normalization). When $o_r \neq 0$, the tower is obstructed at degree r.

Proposition 3.7 (Obstructions vanish for Koszul algebras). *If $\mathcal{A}$ is chirally Koszul, then $o_r(\mathcal{A}) = 0$ for all $r \geq 2$: the tower is unobstructed, and the full MC element $\Theta_{\mathcal{A}}$ exists.*

Proof. The Koszul property implies that the bar cohomology is concentrated in bar degree 1, which forces $H^2(\mathcal{A}^{\mathrm{sh}}_{r,0}) = 0$ for the relevant component of the shadow algebra. The vanishing of the second cohomology group means that every cocycle is a coboundary, so the obstruction class vanishes.

More precisely: the Koszulness of $\mathcal{A}$ implies that the spectral sequence computing the shadow algebra cohomology degenerates at E_1, and the degeneration forces the higher obstructions to be trivial. This is the vertex-algebraic analogue of the fact that a Koszul algebra has a linear resolution: the bar differential has no higher corrections. □

Remark 3.8 (Beyond the Koszul locus). All standard families (Heisenberg, affine Kac–Moody at non-critical level, $\beta\gamma$, Virasoro, $\mathcal{W}_N$) are chirally Koszul, and for these the shadow tower is unobstructed. The obstruction theory becomes nontrivial beyond the Koszul locus: for example, at the critical level $k = -h^\vee$ of affine Kac–Moody, $\kappa = 0$ and the propagator $P = 1/\kappa$ is undefined. The shadow tower does not exist in the standard sense; the modular convolution algebra requires a different filtration (the critical filtration, in which the center of the completed enveloping algebra replaces the curvature κ). Admissible-level quotients $L_k(\mathfrak{g})$ (where null vectors modify the bar complex) are another natural class where obstructions may arise.

3.5. **The shadow algebra structure.** The shadow algebra $\mathcal{A}^{\mathrm{sh}} = H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}))$ carries a graded Lie bracket $[-,-]^{\mathrm{sh}}$, inherited from the graph composition bracket (2.1). The shadow obstruction tower $\{S_r\}$ is a sequence of elements in $\mathcal{A}^{\mathrm{sh}}$ satisfying the projected MC equation.

Proposition 3.9 (Structure of the shadow algebra)**.**

(i) The shadow algebra $\mathcal{A}^{\mathrm{sh}}$ is a pronilpotent graded Lie algebra with degree filtration $F^r\mathcal{A}^{\mathrm{sh}} = \{\text{elements of degree} \geq r\}$.

(ii) On a primary line L, the shadow algebra restricts to a one-dimensional Lie algebra at each degree. The bracket $[\mathrm{Sh}_j, \mathrm{Sh}_k]^{\mathrm{sh}}|_L$ is a scalar multiple of $\mathrm{Sh}_{j+k-2}|_L$, with the scalar determined by the propagator $P = 1/\kappa$ and the symmetry factors.

(iii) The MC equation in $\mathcal{A}^{\mathrm{sh}}$ is equivalent to the algebraic relation $H(t)^2 = t^4\,Q_L(t)$ (Theorem 4.1).

(iv) The shadow algebra carries a $\mathbb{Z}$-grading (by degree) and a $\mathbb{Q}(c)$-rational structure (all structure constants are rational functions of the central charge).

Remark 3.10 (The shadow algebra is not a ring)*.* The shadow algebra $\mathcal{A}^{\mathrm{sh}}$ has a graded Lie bracket, not a ring structure. The MC element $\Theta_{\mathcal{A}}$ is an element of the Lie algebra, not a generator of a ring. The algebraicity theorem (Theorem 4.1) is a statement about the Lie algebra structure: the MC equation, which is quadratic in the Lie bracket, reduces to a quadratic algebraic equation for the generating function.

3.6. **The Heisenberg shadow tower: the base computation.** The Heisenberg algebra is the simplest nontrivial example. Its shadow tower terminates immediately, providing the base case from which all other computations depart.

Example 3.11 (Heisenberg: degree-by-degree construction)**.** For $\mathcal{H}_k$ with $k \neq 0$:

Degree 2*.* The Heisenberg OPE $J(z)J(w) \sim k/(z-w)^2$ is purely quadratic. The collision residue along $d\log(z-w)$ extracts the coefficient of $(z-w)^{-2}$ after absorbing one power: $\mathrm{Res}_{z\to w}\, k/(z-w)^2 \cdot d\log(z-w) = \mathrm{Res}_{z\to w}\, k/(z-w)^2 \cdot dz/(z-w) = k \cdot \mathrm{Res}_{z\to w}(z-w)^{-3}\,dz$, but the bar differential extracts the OPE residue in the standard normalization, giving $S_2 = \kappa = k$.

Degree 3*.* The three-point collision residue on $\mathrm{FM}_3(X)$ involves the OPE $J(z_1)J(z_2)J(z_3)$. Since the Heisenberg OPE has no simple-pole term ($J_{(0)}J = 0$), the cubic collision vanishes: $S_3 = \alpha = 0$.

Degree 4 *and higher.* All higher collision residues vanish for the same reason: the Heisenberg OPE is purely quadratic, so no connected correlator on $\mathrm{FM}_n(X)$ with $n \geq 3$ is nonzero. The shadow metric is $Q_L(t) = 4k^2$, a perfect square, and $H(t) = 2k\,t^2$.

Verification. $H(t)^2 = 4k^2t^4 = t^4 \cdot 4k^2 = t^4 Q_L(t)$. The algebraicity theorem is trivially satisfied.

4. The Riccati algebraicity theorem

4.1. **Statement.**

Theorem 4.1 (Riccati algebraicity)**.** *Let $\mathcal{A}$ be a chirally Koszul vertex algebra, L a primary line with shadow data S_r, curvature $\kappa = S_2 \neq 0$, and cubic coefficient $\alpha = S_3$. Define the shadow metric $Q_L(t)$ and the weighted shadow generating function $H(t) := \sum_{r\geq 2} r\,S_r\,t^r$. Then*

$$H(t)^2 \;=\; t^4\,Q_L(t).$$

Equivalently, $H(t) = t^2\sqrt{Q_L(t)}$. Every shadow coefficient S_r with $r \geq 5$ is determined by the three genus-0 invariants (κ, α, S_4) via $S_r = \frac{1}{r}\,[t^{r-2}]\sqrt{Q_L(t)}$.

4.2. **The single-line recursion.**

Lemma 4.2 (MC recursion on a primary line). *Let $P := 1/\kappa$. On the primary line L, the master equation reduces to*

$$S_r \;=\; -\frac{P}{2r} \sum_{\substack{j+k=r+2 \\ 3\le j\le k}} c_{jk}\, jk\, S_j\, S_k, \qquad c_{jk} = \begin{cases} 1 & j<k, \\ 1/2 & j=k, \end{cases}$$

for all $r \ge 5$.

Proof. On the one-dimensional subspace L, the bracket $[\mathrm{Sh}_j, \mathrm{Sh}_k]$ restricts to a scalar pairing. The cyclic coderivation bracket on L evaluates as $[\mathrm{Sh}_j, \mathrm{Sh}_k]_L = c_{jk}\, jk\, P\, S_j S_k\, x^{j+k-2}$: the factor jk comes from the number of ways to select one input from each coderivation, the factor $P = 1/\kappa$ from the propagator on the internal edge, and c_{jk} corrects for the symmetry when $j = k$. The linear term $d_0(\mathrm{Sh}_r)$ contributes $2r\,\kappa\, S_r$. Dividing by $2r\,\kappa$ gives the stated recursion. □

4.3. **Proof of algebraicity.**

Proof of Theorem 4.1. Define $a_n := (n+2)\, S_{n+2}$ for $n \ge 0$, so that $a_0 = 2\kappa$, $a_1 = 3\alpha$, $a_2 = 4S_4$, and $H(t) = \sum_{n\ge 0} a_n\, t^{n+2}$. Set $F(t) := H(t)/t^2 = \sum_{n\ge 0} a_n\, t^n$. The claim is equivalent to $F(t)^2 = Q_L(t)$.

Expand $F(t)^2 = \sum_{m\ge 0} \big(\sum_{i+j=m} a_i a_j\big)\, t^m$. The first three coefficients are:

$$\begin{aligned} m = 0: &\qquad a_0^2 = 4\kappa^2, \\ m = 1: &\qquad 2a_0 a_1 = 12\kappa\alpha, \\ m = 2: &\qquad a_1^2 + 2a_0 a_2 = 9\alpha^2 + 16\kappa S_4. \end{aligned}$$

These are exactly the coefficients of $Q_L(t)$.

For $m \ge 3$, we show that the convolution $\sum_{i+j=m} a_i a_j = 0$. Setting $r = m+2$:

$$\sum_{i+j=m} a_i a_j \;=\; \sum_{\substack{p+q=r+2 \\ p,q\ge 2}} pq\, S_p\, S_q,$$

where $p = i+2$, $q = j+2$. Splitting off the $p = 2$ and $q = 2$ terms:

$$\sum_{i+j=m} a_i a_j \;=\; 4r\,\kappa\, S_r \;+\; \sum_{\substack{p+q=r+2 \\ p,q\ge 3}} pq\, S_p\, S_q. \tag{4.1}$$

The MC recursion (Lemma 4.2) gives

$$S_r = -\frac{1}{2r\kappa} \sum_{\substack{p+q=r+2 \\ 3\le p\le q}} c_{pq}\, pq\, S_p\, S_q.$$

Symmetrizing: $\sum_{\substack{p+q=r+2 \\ p,q\ge 3}} pq\, S_p\, S_q = 2\sum_{\substack{p+q=r+2 \\ 3\le p\le q}} c_{pq}\, pq\, S_p\, S_q = -4r\,\kappa\, S_r$. Substituting into (4.1):

$$\sum_{i+j=m} a_i a_j \;=\; 4r\,\kappa\, S_r + (-4r\,\kappa\, S_r) \;=\; 0.$$

Hence $[t^m](F^2 - Q_L) = 0$ for all $m \ge 3$. Combined with the verification at $m = 0, 1, 2$, this proves $F(t)^2 = Q_L(t)$.

The closed form follows: $r\, S_r = [t^{r-2}]F(t) = [t^{r-2}]\sqrt{Q_L(t)}$. □

Remark 4.3 (Riccati ODE). Set $U(t) := \sum_{r\ge 3} S_r\, t^r$, the nonlinear part of the shadow generating function. The algebraic relation is equivalent to the first-order ODE

$$2t\, U'(t) \;+\; \frac{1}{2\kappa}\big(U'(t)\big)^2 \;=\; 6\alpha\, t^3 + \frac{9\alpha^2 + 2\Delta}{2\kappa}\, t^4,$$

quadratic in U': the shadow obstruction tower satisfies a Riccati-type equation of degree 2, and all terms at t^r for $r \geq 5$ cancel identically.

Remark 4.4 (The intrinsic quartic principle). Three numbers determine an infinite sequence. The analogue is the Weierstrass equation: two parameters (g_2, g_3) determine all Laurent coefficients of $\wp$, because $\wp$ satisfies an algebraic equation of degree 2 in $\wp'$. Here three parameters (κ, α, S_4) determine all S_r, because H satisfies an algebraic equation of degree 2.

4.4. **Closed-form extraction of shadow coefficients.** The algebraicity theorem converts the recursive computation of shadow coefficients into a single binomial expansion.

Corollary 4.5 (Explicit extraction formula). *For $r \geq 2$, the shadow coefficient is*

$$S_r \;=\; \frac{1}{r}\,[t^{r-2}]\sqrt{Q_L(t)} \;=\; \frac{1}{r}\,[t^{r-2}]\sqrt{4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2}.$$

Expanding the square root via the binomial series in the normalized variable $u := (3\alpha\, t)/(2\kappa) + (9\alpha^2 + 2\Delta)\, t^2/(4\kappa^2)$:

$$\sqrt{Q_L(t)} \;=\; 2\kappa \sum_{n=0}^{\infty} \binom{1/2}{n} u^n,$$

where $\binom{1/2}{n} = \frac{(1/2)(1/2-1)\cdots(1/2-n+1)}{n!} = \frac{(-1)^{n-1}}{2^{2n-1}\,n}\binom{2n-2}{n-1}$ for $n \geq 1$.

Example 4.6 (First five coefficients by extraction). Writing $a := 3\alpha/(2\kappa)$ and $b := (9\alpha^2 + 2\Delta)/(4\kappa^2)$ for the normalized coefficients of u:

$$\begin{aligned}
S_2 &= \kappa,\\
S_3 &= \alpha,\\
S_4 &= S_4 \qquad \text{(tautological at this degree)},\\
S_5 &= \frac{2\kappa}{5}\left(-\frac{a\,b}{4} + \frac{a^3}{16}\right) \;=\; -\frac{3\alpha\,\Delta}{10\,\kappa^2},\\
S_6 &= \frac{2\kappa}{6}\left(\frac{b^2}{16} - \frac{a^2 b}{8} + \frac{5a^4}{128}\right) \;=\; \frac{\Delta\,(45\alpha^2 + 2\Delta)}{60\,\kappa^3}.
\end{aligned}$$

The key structural observation: every coefficient S_r with $r \geq 4$ contains Δ as a factor. This is the universal factorization property of Theorem 5.1.

Remark 4.7 (Comparison with direct recursion). The MC recursion (Lemma 4.2) computes S_r as a sum of $\lfloor (r-2)/2 \rfloor$ terms, each involving products of previously computed coefficients. The closed-form extraction $S_r = \frac{1}{r}[t^{r-2}]\sqrt{Q_L}$ replaces this recursive computation with a single coefficient extraction from a degree-2 algebraic function. The computational advantage grows with r: at degree 10, the recursion involves 4 nonlinear terms, while the extraction requires expanding $\sqrt{1+u}$ to order 8.

For computer algebra, both methods are efficient. The conceptual advantage of the closed form is structural: it makes the Δ-divisibility and the asymptotic behaviour manifest, and it identifies the shadow tower as a branch of an algebraic curve rather than as the output of a recursion.

4.5. **The spectral curve.** The algebraicity theorem identifies the shadow obstruction tower with a point on an algebraic curve.

Definition 4.8 (Shadow spectral curve). The *shadow spectral curve* of a primary line L with shadow metric Q_L is the affine curve

$$\Sigma_L \;:=\; \big\{(t, y) \in \mathbb{A}^2 : y^2 = Q_L(t)\big\}.$$

The shadow generating function $H(t) = t^2 y$ selects one of the two branches of Σ_L near $t = 0$ (the branch with $y(0) = 2\kappa > 0$).

For class **G**, $Q_L = (2\kappa)^2$ is constant and Σ_L degenerates to two horizontal lines. For class **L**, Q_L has a double zero at $t = -2\kappa/(3\alpha)$ and Σ_L is a rational curve with a node. For class **M**, Q_L has two distinct zeros, Σ_L is a smooth affine conic (genus 0), and the two branches of $\sqrt{Q_L}$ are interchanged by monodromy around the branch points.

Remark 4.9 (Comparison with matrix models). In the Eynard–Orantin topological recursion, the spectral curve is prescribed as input and determines the full genus expansion. Here the situation is reversed: the vertex algebra determines the shadow metric Q_L (three numbers), which determines the spectral curve Σ_L, which determines the shadow tower. The spectral curve is an *output*, not an input.

The structural parallel is precise: the recursion kernel of Eynard–Orantin is the Bergman kernel on the spectral curve; the MC recursion of the shadow tower is the convolution kernel on the parameter line, and the algebraicity theorem identifies the two. The genus-g free energies F_g are computed from the shadow tower by stable graph summation, paralleling the Feynman expansion from the spectral curve in matrix models. The key difference is that the shadow spectral curve has genus 0 (it is a conic), while matrix-model spectral curves may have arbitrary genus.

5. The four-class classification

5.1. The single-line dichotomy.

Theorem 5.1 (Single-line dichotomy). *On any primary line L with $\kappa \neq 0$ and critical discriminant $\Delta = 8\kappa S_4$, the shadow depth satisfies $r_{\max} \in \{2, 3, \infty\}$:*

*(i) $\Delta = 0$, $\alpha = 0$: $Q_L = (2\kappa)^2$; $H(t) = 2\kappa\, t^2$; $r_{\max} = 2$ (class **G**, Gaussian).*
*(ii) $\Delta = 0$, $\alpha \neq 0$: $Q_L = (2\kappa + 3\alpha\, t)^2$; $H(t) = t^2(2\kappa + 3\alpha\, t)$; $r_{\max} = 3$ (class **L**, Lie).*
*(iii) $\Delta \neq 0$: Q_L is irreducible; $S_r \neq 0$ for all $r \geq 4$ generically; $r_{\max} = \infty$ (class **M**, mixed).*

For $r \geq 4$: $S_r = \Delta \cdot R_r$ with $R_r \in \mathbb{Q}(\alpha, \kappa, S_4)$ a universal rational function independent of the algebra family.

Proof. The Gaussian decomposition $Q_L(t) = (2\kappa + 3\alpha\, t)^2 + 2\Delta\, t^2$ shows that Q_L is a perfect square if and only if $\Delta = 0$.

Case $\Delta = 0$. Then $Q_L = (2\kappa + 3\alpha\, t)^2$, so $H(t) = t^2(2\kappa + 3\alpha\, t)$. If also $\alpha = 0$, then $H(t) = 2\kappa\, t^2$; $S_r = 0$ for all $r \geq 3$ (class **G**). If $\alpha \neq 0$, then H is a cubic polynomial; $S_r = 0$ for $r \geq 4$, while $S_3 = \alpha \neq 0$ (class **L**).

Case $\Delta \neq 0$. Then Q_L has two distinct zeros, so $\sqrt{Q_L(t)}$ is not polynomial. The Taylor coefficients $S_r = \frac{1}{r}[t^{r-2}]\sqrt{Q_L(t)}$ are nonzero for all $r \geq 4$ generically (the only way $[t^n]\sqrt{Q_L}$ vanishes identically is if $\sqrt{Q_L}$ is polynomial, requiring $\Delta = 0$). Hence $r_{\max} = \infty$ (class **M**).

For the Δ-divisibility: write $\sqrt{Q_L} = |2\kappa + 3\alpha\, t|\,(1 + 2\Delta\, t^2/(2\kappa + 3\alpha\, t)^2)^{1/2}$ and expand the binomial series. Every term beyond zeroth order contributes a factor of Δ. □

5.2. Class C: contact depth.

A fourth class escapes the single-line dichotomy. The $\beta\gamma$ system exhibits the mechanism concretely.

The $\beta\gamma$ algebra at conformal weight λ has two generators β (weight λ) and γ (weight $1 - \lambda$) with OPE $\beta(z)\gamma(w) \sim 1/(z - w)$: a single simple pole, no self-OPE. The central charge is $c_{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1)$ and the modular characteristic is $\kappa(\beta\gamma) = c_{\beta\gamma}/2 = 6\lambda^2 - 6\lambda + 1$. At $\lambda = 1$: $\kappa = 1$, $c = 2$.

The r-matrix is the constant permutation $r(z) = P/z$ (the transposition of β and γ channels): a simple pole, one order below the OPE double pole (the $d\log$ propagator absorbs one pole as always). The cubic shadow $S_3 = 0$: both β and γ are free fields with no self-coupling, so the Lie bracket component vanishes. The discriminant $\Delta = 8\kappa S_4$: since $S_3 = 0$, the single-line dichotomy gives $\Delta = 0$ on every one-dimensional primary line. The single-line tower terminates at degree 2.

The quartic contact invariant $S_4 \neq 0$ lives on the *two-field* stratum: the cross-channel $\beta\gamma\beta\gamma$ four-point amplitude produces a nonzero quartic residue from the interaction of two distinct fields. Concretely, the degree-4 MC component $\Theta^{(4)}_{\beta\gamma}$ has a nonvanishing contribution from the graph with two $\beta\gamma$ propagators crossing; this contribution is proportional to $(1-2\lambda)^2$ and vanishes only at $\lambda = 1/2$ (the symplectic boson, where $c = -1$ and the theory acquires enhanced symmetry). The quintic and all higher obstructions vanish: the cross-channel self-bracket exits the two-field complex by dimensional constraint (a rank-one target cannot support a quintic interaction with the given conformal weight pairing).

Definition 5.2 (Class **C**). A chirally Koszul vertex algebra is of *class* **C** (contact, $r_{\max} = 4$, $d_{\mathrm{alg}} = 2$) if:

(a) on every one-dimensional primary line L, the shadow metric Q_L is a perfect square ($\Delta_L = 0$), so the single-line tower terminates;
(b) in the full multi-field deformation complex, a nonzero quartic contact invariant $S_4^{\mathrm{cross}} \neq 0$ arises from the cross-channel interaction of distinct generators at complementary conformal weights;
(c) the quintic and all higher obstructions vanish by dimensional constraint on the cross-channel complex.

The $\beta\gamma$ system at generic $\lambda \neq 1/2$ is the archetype: $S_3 = 0$ (no Lie bracket), $S_4 \neq 0$ (quartic contact from $\beta\gamma$ crossing), $S_r = 0$ for $r \geq 5$ (dimensional termination). The algebraic depth is $d_{\mathrm{alg}} = 2$: finite but nonzero, distinguishing class **C** from both class **G** ($d_{\mathrm{alg}} = 0$) and class **L** ($d_{\mathrm{alg}} = 1$).

The four classes partition the Koszul world:

Class	α	Δ	$r_{\max}$	Examples
G	0	0	2	Heisenberg, lattice, free fermion
L	$\neq 0$	0	3	Affine Kac–Moody (all types)
C	0^*	0^*	4	$\beta\gamma$, bc ghosts
M	$\neq 0$	$\neq 0$	∞	Virasoro, $\mathcal{W}_N$ ($N \geq 3$)

All four classes are Koszul: shadow depth classifies the complexity of the OPE within the Koszul world, not Koszulness itself.

6. The depth gap theorem

6.1. Statement and proof.

Definition 6.1 (Algebraic depth). The *algebraic depth* of a chirally Koszul vertex algebra $\mathcal{A}$ is

$$d_{\mathrm{alg}}(\mathcal{A}) \;:=\; \begin{cases} 0 & \text{if } S_r = 0 \text{ for all } r \geq 3 \quad (\text{class } \mathbf{G}), \\ 1 & \text{if } \alpha \neq 0 \text{ and } S_r = 0 \text{ for all } r \geq 4 \quad (\text{class } \mathbf{L}), \\ 2 & \text{if } r_{\max} = 4 \quad (\text{class } \mathbf{C}), \\ \infty & \text{if } r_{\max} = \infty \quad (\text{class } \mathbf{M}). \end{cases}$$

Proposition 6.2 (Depth gap). *Let $\mathcal{A}$ be a chirally Koszul algebra in the standard landscape. Then*

$$d_{\mathrm{alg}}(\mathcal{A}) \;\in\; \{0,\, 1,\, 2,\, \infty\}.$$

No value 3 or any finite value ≥ 3 is realized. The four permitted values correspond bijectively to the classes **G**, **L**, **C**, **M**.

Proof. The argument has two parts: the single-line Riccati regime, and the global contact witness.

Single-line regime ($\kappa \neq 0$). On a primary line L with $\kappa \neq 0$, the shadow generating function $H(t) = t^2\sqrt{Q_L(t)}$ is determined by (κ, α, S_4) (Theorem 4.1), where $Q_L(t) = (2\kappa + 3\alpha\, t)^2 + 2\Delta\, t^2$.

If $\Delta = 0$ (equivalently $S_4 = 0$), then $Q_L = (2\kappa + 3\alpha\, t)^2$ is a perfect square, so $H(t) = t^2(2\kappa + 3\alpha\, t)$ is a polynomial of degree 3. The shadow coefficients $S_r = 0$ for all $r \geq 4$. The tower terminates at degree at most 3, giving $d_{\mathrm{alg}} \in \{0, 1\}$.

If $\Delta \neq 0$, then Q_L is not a perfect square. The binomial series for $(1 + u)^{1/2}$ with $u = (6\alpha\, t + (9\alpha^2 + 2\Delta)\, t^2)/(4\kappa^2)$ does not terminate when the coefficient of t^2 in u is nonzero. Therefore $S_r \neq 0$ for infinitely many r, and $d_{\mathrm{alg}} = \infty$.

No single primary line with $\kappa \neq 0$ can realize $d_{\mathrm{alg}} = 2$ or 3: the sharp dichotomy $\Delta = 0 \Rightarrow$ polynomial, $\Delta \neq 0 \Rightarrow$ infinite, admits no intermediate depth.

The degree-4 master equation isolates the quartic cancellation:

$$S_4^{\mathrm{free}} \;=\; -\frac{9\alpha^2}{16\kappa},$$

so the finite-depth locus at degree 4 is exactly $S_4 = 0$, the Jacobi locus. Once this cancellation fails, the next recursion step forces a tail. At $r = 5$:

$$S_5 = -\frac{6}{5}\, P\, \alpha\, S_4.$$

For $\kappa \neq 0$, $\alpha \neq 0$, and $S_4 \neq 0$, this is nonzero. If $\alpha = 0$ but $S_4 \neq 0$, then $S_5 = 0$ but

$$S_6 \;=\; -\frac{P}{12} \cdot \frac{1}{2} \cdot 16\, S_4^2 \;=\; -\frac{2}{3}\, P\, S_4^2 \;\neq\; 0.$$

Either way, the tower does not terminate at degree 5.

Global contact witness. The value $d_{\mathrm{alg}} = 2$ is realized by the $\beta\gamma$ system (equivalently bc), whose global shadow data on the conformal-weight family is $S_2 = 6\lambda^2 - 6\lambda + 1$, $S_3 = 0$, $S_4 = -5/12$, $S_r = 0$ for $r \geq 5$. The internal weight-changing line is shadow-trivial, so the class-**C** witness is the standard conformal-weight family itself (Definition 5.2). □

Remark 6.3 (Why no $d_{\mathrm{alg}} = 3$). The impossibility of $d_{\mathrm{alg}} = 3$ has a clean algebraic explanation. On a single primary line, the tower is controlled by a degree-2 algebraic function: either the square root is rational (tower terminates at degree ≤ 3) or irrational (tower infinite). There is no mechanism for a degree-2 algebraic function to produce exactly one nonzero coefficient beyond the polynomial envelope.

The standard conformal-weight family $\beta\gamma_\lambda$ (equivalently bc_λ) supplies the unique finite nonlinear case beyond classes **G** and **L**: $S_3 = 0$, $S_4 = -5/12$, and $S_r = 0$ for $r \geq 5$. Hence d_{alg} can equal 2, but no finite value greater than 2 occurs.

Remark 6.4 (Generating depth vs algebraic depth). The algebraic depth d_{alg} (whether the tower terminates) is distinct from the generating depth d_{gen} (the degree at which all higher operations are determined by lower ones). For Virasoro: $d_{\mathrm{gen}} = 3$ (the cubic shadow $\alpha = 2$ generates all higher shadows recursively from (κ, α, S_4)), but $d_{\mathrm{alg}} = \infty$ (all S_r are nonzero).

6.2. **The Jacobi mechanism.**

The vanishing $S_4 = 0$ for class **L** has a precise algebraic explanation that illuminates the depth gap.

Proposition 6.5 (Jacobi kills the quartic shadow). *Let $\mathcal{A} = V_k(\mathfrak{g})$ be an affine Kac–Moody vertex algebra. The quartic collision residue on* $\mathrm{FM}_4(X)$*, restricted to the primary line, is proportional to*

$$\mathcal{J} \;:=\; \sum_{\mathrm{cyclic}(a,b,c)} f^{abx} f^{xcd},$$

where f^{abc} are the structure constants of $\mathfrak{g}$. The Jacobi identity $[[J^a, J^b], J^c] + \text{cyclic} = 0$ forces $\mathcal{J} = 0$, hence $S_4 = 0$.

Proof. The degree-4 component of the MC element $\Theta_{\mathcal{A}}$ receives contributions from trees with one internal edge connecting two cubic vertices. Each cubic vertex contributes f^{abc}; the internal edge carries the propagator $P = 1/\kappa$; and the contraction over the internal index x produces $f^{abx}f^{xcd}$. The master equation at degree 4 reads $d_0(\mathrm{Sh}_4) + [\mathrm{Sh}_3, \mathrm{Sh}_3] = 0$.

The bracket $[\mathrm{Sh}_3, \mathrm{Sh}_3]$ evaluates on the primary line as a sum over channel pairings, and the coefficient of S_4 in the linear term $d_0(\mathrm{Sh}_4)$ is $8\kappa S_4$. The obstruction is therefore $8\kappa S_4 = -[\mathrm{Sh}_3, \mathrm{Sh}_3]|_{\deg 4}$, which is proportional to $\mathcal{J}$. Since $\mathcal{J} = 0$ by the Jacobi identity, $S_4 = 0$. □

Remark 6.6 (The Jacobi locus as a discriminant). The vanishing of S_4 is not a numerical accident: it is the statement that the Jacobi identity of the Lie algebra $\mathfrak{g}$ kills the quartic shadow. This explains the structural difference between classes **L** and **M**: affine Kac–Moody algebras carry a Lie bracket whose Jacobi identity kills S_4; Virasoro and $\mathcal{W}_N$ algebras carry OPE structures without a classical Jacobi identity, and S_4 survives.

The Jacobi locus $\{S_4 = 0\} \subset \{\text{shadow data}\}$ is a codimension-1 subvariety of the parameter space. On this locus, $\Delta = 0$ and the tower terminates. Off the locus, $\Delta \neq 0$ and the tower is infinite. The transition is sharp: the depth jumps from 3 to ∞ with no intermediate value.

6.3. **The shadow Lie algebra perspective.** The depth gap admits a second proof through the structure of the shadow algebra $\mathcal{A}^{\mathrm{sh}} := H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}))$.

Proposition 6.7 (Shadow Lie algebra Jacobi). *The shadow algebra $\mathcal{A}^{\mathrm{sh}}$ carries a graded Lie bracket $[-,-]^{\mathrm{sh}}$. At degree 4, the Jacobi identity in $\mathcal{A}^{\mathrm{sh}}$ forces*

$$[S_3, [S_3, S_3]^{\mathrm{sh}}]^{\mathrm{sh}} = 0,$$

and the degree-5 master equation reads

$$d_0(\mathrm{Sh}_5) + [S_3, S_4]^{\mathrm{sh}} + \tfrac{1}{2}[S_4, S_3]^{\mathrm{sh}} = 0.$$

If $S_4 \neq 0$ and $S_3 \neq 0$, then $S_5 \neq 0$; if $S_4 \neq 0$ and $S_3 = 0$, then $S_6 \neq 0$. In either case, the tower does not terminate at degree 5.

This is the shadow Lie algebra Jacobi mechanism identified in the monograph (Proposition 6.2, alternative proof): the graded Jacobi identity in the shadow algebra constrains which depths are achievable. The depth gap $d_{\mathrm{alg}} \neq 3$ is forced because no configuration of (S_3, S_4) with $S_4 \neq 0$ can produce $S_5 = S_6 = \cdots = 0$.

7. SC-formality characterizes class G

The Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$ is a two-coloured operad governing the interaction of a bulk algebra (closed colour) with a boundary algebra (open colour). A chirally Koszul vertex algebra $\mathcal{A}$ is *SC-formal* if the transferred Swiss-cheese operations m_k^{SC} vanish for all $k \geq 3$.

Proposition 7.1 (SC-formality iff class **G**). *Let $\mathcal{A}$ be a chirally Koszul vertex algebra in the standard landscape. Then $\mathcal{A}$ is SC-formal if and only if $\mathcal{A}$ belongs to class* **G**.

Proof. Forward direction (class **G** $\Longrightarrow$ *SC-formal).* Class **G** consists of the Gaussian families: Heisenberg, lattice VOAs, and free fermions. In each case, the genus-0 mixed tree formulas stop at degree 2 (the OPE is purely quadratic), so $m_k^{\mathrm{SC}} = 0$ for $k \geq 3$. The argument uses Wick factorization: all higher connected contractions vanish for a free field.

Converse (SC-formal $\Longrightarrow$ class **G***).* Suppose $\mathcal{A}$ is SC-formal. Then every mixed SC operation of degree $r \geq 3$ vanishes. The tree-shadow correspondence is operadic: the degree-r mixed tree and the degree-r shadow are produced by the same genus-0 tree-transfer formula, with the same propagator on internal edges. The only difference is the output colour, and passage to the symmetric closed sector applies the averaging morphism $\mathrm{av}(\Theta_{\mathcal{A}}^{E_1}) = \Theta_{\mathcal{A}}$.

Hence SC-formality forces $S_r(\mathcal{A}) = 0$ for every $r \geq 3$: a nonzero shadow coefficient would produce, by the tree-shadow correspondence, a nontrivial mixed degree-r Massey product, contradicting SC-formality.

The condition $S_r = 0$ for all $r \geq 3$ is exactly class **G** (Theorem 5.1(i)). Classes **L**, **C**, and **M** are ruled out: class **L** has $S_3 = \alpha \neq 0$; class **C** has a nonzero quartic contact invariant; class **M** has $S_r \neq 0$ for all $r \geq 4$. □

Remark 7.2 (Operadic alternative). The proposition admits a second proof via genus-0 operadic transfer. Class **G** is the Gaussian locus where the genus-0 transfer is generated by the binary two-point kernel alone. Every connected SC tree of degree $r \geq 3$ factors through binary pairings and carries no primitive higher vertex, so the mixed operations vanish. For the converse, if some $S_r \neq 0$ with $r \geq 3$, the shadow-formality identification and the averaging identity $\mathrm{av}(\Theta_{\mathcal{A}}^{E_1}) = \Theta_{\mathcal{A}}$ exhibit a nonzero closed projection from a nonzero tree, contradicting SC-formality.

7.1. The shadow-formality identification.

The shadow obstruction tower and the A_∞ non-formality data of a chirally Koszul algebra are identified by the homotopy transfer theorem.

Proposition 7.3 (Shadow-formality identification). *Let $\mathcal{A}$ be a chirally Koszul vertex algebra and $\{m_k\}_{k\geq 2}$ the transferred A_∞ structure on the bar cohomology $H^\bullet(B(\mathcal{A}))$, obtained by homological perturbation from the bar differential. Then:*

(i) The degree-2 shadow $S_2 = \kappa$ corresponds to the binary product m_2 (which is always nonzero for $\kappa \neq 0$).

(ii) The degree-3 shadow $S_3 = \alpha$ corresponds to the first A_∞ correction m_3: $\alpha = 0$ if and only if $m_3 = 0$ on the primary line.

(iii) The degree-r shadow S_r corresponds to m_r: $S_r = 0$ if and only if $m_r = 0$ on the primary line (after projection from the full A_∞ structure).

(iv) The shadow depth d_{alg} equals the A_∞-depth: the minimal n such that $m_k = 0$ for all $k > n + 2$ on the primary line.

Proof. The homotopy transfer theorem produces the A_∞ structure $\{m_k\}$ by summing over binary trees with k leaves, where each internal edge carries the homotopy $h = P \cdot d_0^{-1}$ (with $P = 1/\kappa$ the propagator) and each internal vertex carries the binary product m_2. The degree-r component of the MC element $\Theta_{\mathcal{A}}$ is the generating function of the tree sum at degree r. The shadow coefficient S_r is the scalar restriction of this tree sum to the primary line.

The identification $S_r \leftrightarrow m_r|_L$ follows from the fact that the MC recursion (Proposition 3.2) and the A_∞ recursion (the Stasheff identities) are the same system of equations, restricted to different components of the deformation complex. On the primary line, both reduce to the scalar recursion of Lemma 4.2. □

Corollary 7.4 (Shadow depth classifies A_∞-formality). *The four shadow depth classes correspond to four levels of A_∞-formality:*

Class	d_{alg}	A_∞ *formality*	*Status*
G	0	*formal* ($m_k = 0$, $k \geq 3$)	*SC-formal*
L	1	$m_3 \neq 0$, $m_k = 0$ *for* $k \geq 4$	*1-formal*
C	2	$m_4 \neq 0$ *on standard conformal-weight family*	*2-formal*
M	∞	$m_k \neq 0$ *for all* k	*non-formal*

The depth gap $d_{\mathrm{alg}} \neq 3$ *translates to: no chirally Koszul vertex algebra has* A_∞*-depth exactly* 3 *(i.e.,* $m_3 \neq 0$, $m_4 \neq 0$, *but* $m_5 = m_6 = \cdots = 0$*).*

Remark 7.5 (The L_∞ perspective). The modular convolution algebra $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$ is a dg Lie algebra, and the MC element $\Theta_{\mathcal{A}}$ determines a gauge equivalence class. The shadow-formality identification has an L_∞ counterpart: the transferred L_∞ structure $\{\ell_k\}_{k\geq 2}$ on the shadow algebra $\mathcal{A}^{\mathrm{sh}}$ satisfies $\ell_k = 0$ for $k \geq 3$ if and only if $\mathcal{A}$ is class **G**. The L_∞ formality level coincides with the A_∞-depth on the primary line.

The passage from A_∞ to L_∞ is the passage from the open colour (boundary, A_∞) to the closed colour (bulk, L_∞) in the Swiss-cheese structure. SC-formality is the simultaneous vanishing of both the A_∞ and L_∞ higher operations, which occurs exactly in class **G** (Proposition 7.1).

Remark 7.6 (Massey products). The A_∞ correction m_3 is a Massey product: it measures the failure of the binary product to be associative up to homotopy. For class **L** (affine Kac–Moody), $m_3 \neq 0$ because the Lie bracket introduces a cubic non-associativity. For class **G** (Heisenberg), $m_3 = 0$ because the abelian OPE is strictly associative.

The quartic correction m_4 is a higher Massey product, and its non-vanishing for class **M** is the statement that the Virasoro OPE has an irreducible four-point contact term not generated by lower operations. The depth gap theorem says that this contact term, once present, forces an infinite sequence of higher Massey products: no finite truncation is possible beyond degree 4.

8. The E_1 framing

The shadow obstruction tower is the Σ_n-coinvariant projection of a richer E_1 structure.

Proposition 8.1 (E_1 framing of the shadow tower)**.** *The shadow obstruction tower* $\{S_r\}_{r\geq 2}$ *is the image of the ordered* R*-matrix tower under the averaging map* $\mathrm{av}\colon \mathfrak{g}^{E_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$*. At degree 2:*

- *(i) For abelian algebras (Heisenberg, free fields):* $\kappa = \mathrm{av}(r(z))$*, where* $r^{\mathrm{Heis}}(z) = k/z$ *(level prefix* k *mandatory; at* $k = 0$*, the* r*-matrix vanishes).*
- *(ii) For non-abelian affine Kac–Moody* $V_k(\mathfrak{g})$ *(trace-form convention* $r(z) = k\Omega/z$*, where* Ω *is the inverse Killing form Casimir):*

$$\kappa(V_k(\mathfrak{g})) \;=\; \mathrm{av}(r(z)) + \frac{\dim(\mathfrak{g})}{2} \;=\; \frac{k\,\dim(\mathfrak{g})}{2h^\vee} + \frac{\dim(\mathfrak{g})}{2} \;=\; \frac{\dim(\mathfrak{g})\,(k+h^\vee)}{2h^\vee}.$$

The $\dim(\mathfrak{g})/2$ *term is the Sugawara shift, the simple-pole self-contraction through the adjoint Casimir eigenvalue* $2h^\vee$.

At higher degrees, the averaging map sends the degree-r *component of the ordered* R*-matrix tower* $\Theta^{E_1}_{\mathcal{A}}$ *to the shadow* S_r*.*

Proof. The ordered bar $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ carries the E_1 convolution algebra $\mathfrak{g}^{E_1}_{\mathcal{A}}$, whose MC element $\Theta^{E_1}_{\mathcal{A}}$ projects to $\Theta_{\mathcal{A}}$ under av. The degree-2 component of $\Theta^{E_1}_{\mathcal{A}}$ is the R-matrix $r(z)$, and its average is the scalar degree-2 shadow: it equals the modular characteristic κ on abelian families, while for non-abelian affine Kac–Moody the average is κ_{dp} and the full κ adds the Sugawara correction from the simple-pole self-contraction of the ordered two-point function. The higher degrees follow by the same averaging applied to the degree-r MC components. □

Remark 8.2 (What averaging discards). The averaging map av is lossy: it discards the spectral parameter z, the R-matrix structure, and the Yangian datum. The full E_1 tower retains the braiding of line operators; the shadow tower retains only the scalar invariants. For the Virasoro R-matrix $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$, averaging produces $\kappa = c/2$ and $\alpha = 2$, but discards the cubic pole structure that encodes the braided tensor category of Virasoro modules.

8.1. **The R-matrix tower for each class.** The E_1 convolution algebra $\mathfrak{g}^{E_1}_{\mathcal{A}}$ carries a spectral R-matrix $r(z) \in \mathrm{End}(V \otimes V)((z))$ at degree 2, whose averaging produces the scalar degree-2 shadow: for abelian families this is κ, while for non-abelian affine Kac–Moody it is κ_{dp} and the full modular characteristic is $\kappa = \kappa_{\mathrm{dp}} + \dim(\mathfrak{g})/2$. The structure of $r(z)$ varies by class.

Example 8.3 (Class **G**: Heisenberg)**.** $r^{\mathrm{Heis}}(z) = k/z$. One pole of order 1 (from the double-pole OPE $J(z)J(w) \sim k/(z-w)^2$ after $d\log$ absorption). Averaging: $\mathrm{av}(k/z) = k = \kappa$. The R-matrix is scalar, the Yangian is trivial, and the shadow tower has $\alpha = 0$, $\Delta = 0$. At $k = 0$, the R-matrix vanishes; this is the abelian limit, not the critical level (which does not exist for Heisenberg).

Example 8.4 (Class **L**: affine $\mathfrak{sl}_2$)**.** In the trace-form convention: $r^{\mathrm{KM}}(z) = k\,\Omega/z$, where $\Omega = \sum_a t^a \otimes t_a$ is the inverse Killing form Casimir. Averaging over Σ_2-coinvariants: $\mathrm{av}(k\Omega/z) = k\,\mathrm{tr}(\Omega)/\dim(\mathfrak{g}) \cdot \dim(\mathfrak{g})/(2h^\vee) = k\ \dim(\mathfrak{g})/(2h^\vee)$. The full κ includes the Sugawara shift $\dim(\mathfrak{g})/2$, the simple-pole self-contraction through the adjoint Casimir eigenvalue $2h^\vee$.

For $\mathfrak{g} = \mathfrak{sl}_2$: $\dim(\mathfrak{g}) = 3$, $h^\vee = 2$, $\kappa = 3(k+2)/4$. At $k = 0$: $\kappa = 3/2 \neq 0$ (the Lie bracket persists; the R-matrix $r(z)|_{k=0} = 0$ in trace-form convention, but the Sugawara shift $\dim(\mathfrak{g})/2 = 3/2$ survives). At $k = -2$ (critical): $\kappa = 0$, the center jumps, Koszulness fails.

The cubic shadow α encodes the Lie bracket structure constants: the Killing 3-cocycle $\omega_3(X, Y, Z) = \mathrm{tr}([X, Y]Z)$.

Example 8.5 (Class **M**: Virasoro)**.** The Virasoro OPE $T(z)T(w) \sim (c/2)/(z-w)^4 + 2T(w)/(z-w)^2 + \partial T(w)/(z-w)$ has poles at orders 4, 2, 1. After $d\log$ absorption (which reduces each pole order by 1):

$$r^{\mathrm{Vir}}(z) \;=\; \frac{c/2}{z^3} + \frac{2T}{z}.$$

The cubic pole is characteristic of class **M** and encodes the nonlinear structure of the Virasoro algebra. Averaging: $\mathrm{av}((c/2)/z^3) = 0$ (odd-degree pole averages to zero on the primary line), and $\mathrm{av}(2T/z) = c/2 = \kappa$. The cubic shadow $\alpha = 2$ comes from the simple pole coefficient $2T$, whose projection onto the primary line is the scalar 2.

The quartic shadow $S_4 = 10/[c(5c+22)]$ encodes the four-point contact term in the Virasoro OPE, which is not killed by any Jacobi-type identity. This is the mechanism that produces $\Delta \neq 0$ and the infinite tower.

Remark 8.6 (Pole order and shadow class). The R-matrix pole order determines the shadow class:

OPE pole order p	R-matrix pole $p-1$	Class	Example
2	1	**G**	Heisenberg
2 (non-abelian)	1	**L**	affine KM
1 (charged)	0	**C**	$\beta\gamma$
4	3	**M**	Virasoro

The $d$log absorption shifts all pole orders by -1. The Jacobi identity kills the quartic contribution for class **L** (where the non-abelian cubic OPE comes from a Lie bracket). For class **M**, the quartic OPE survives.

8.2. **The ordered-to-symmetric functor.** The averaging map $\mathrm{av}\colon \mathfrak{g}^{E_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is the Σ_n-coinvariant projection. At the level of generating functions, it sends the spectral R-matrix tower $\Theta^{E_1}_{\mathcal{A}}(z) = \sum_{r\geq 2} \mathrm{Sh}^{E_1}_r(z)$ to the shadow tower $\Theta_{\mathcal{A}} = \sum_{r\geq 2} S_r \cdot x^r$ by discarding the spectral parameter z and averaging over permutations.

Proposition 8.7 (Structure of the averaging map)**.** *The averaging map has the following properties:*

(i) Surjectivity. *Every shadow invariant S_r is in the image of* av*; the ordered tower lifts the symmetric tower.*

(ii) Non-injectivity. *The kernel* $\ker(\mathrm{av})$ *contains all Σ_n-anti-invariant components, including the R-matrix braiding data, the Yangian coproduct corrections, and the non-commutative shadow coefficients at each degree.*

(iii) Compatibility with MC. av *is a morphism of dg Lie algebras:* $\mathrm{av}([\xi,\eta]^{E_1}) = [\mathrm{av}(\xi),\mathrm{av}(\eta)]^{\mathrm{mod}}$. *The MC element projects:* $\mathrm{av}(\Theta^{E_1}_{\mathcal{A}}) = \Theta_{\mathcal{A}}$.

(iv) Degree-2 content. *At degree 2:* $\mathrm{av}(r(z)) = \kappa$ *for abelian algebras;* $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2 = \kappa$ *for non-abelian KM (Sugawara shift).*

Remark 8.8 (What the shadow tower cannot see)*.* The shadow tower is blind to: the spectral parameter z and all z-dependent structure; the R-matrix and its Yang–Baxter equation; the Yangian coproduct; the braided tensor category structure on modules; and the Stokes phenomena of the KZ connection. These richer structures live in the E_1 tower and are invisible to the Σ_n-coinvariant projection. The shadow tower sees only the *scalar invariants* of the E_1 structure: the numbers κ, α, S_4, and their higher-degree extensions.

This is the trade-off of averaging: the shadow tower is computable (finite-dimensional at each degree), algebraic (degree 2), and universal (the same structural theorems hold for all families); but it discards the representation-theoretic content that distinguishes, for example, the Yangian of $\mathfrak{sl}_2$ from the Yangian of $\mathfrak{sl}_3$.

9. The shadow connection and Koszul monodromy

9.1. The shadow connection.

Definition 9.1 (Shadow connection)**.** The *shadow connection* on $L \setminus \{Q_L = 0\}$ is

$$\nabla^{\mathrm{sh}} := d - \frac{Q_L'(t)}{2\,Q_L(t)}\,dt = d - d\Big(\log\sqrt{Q_L}\Big).$$

Proposition 9.2 (Properties)**.**

(i) Flatness. $(\nabla^{\mathrm{sh}})^2 = 0$.

(ii) Regular singularities. *At each zero t_0 of Q_L, the connection form has a simple pole with residue $\frac{1}{2}$.*

(iii) Koszul monodromy. *The monodromy around each branch point is* $\exp(2\pi i \cdot \frac{1}{2}) = -1$.

(iv) Flat section. *The unique flat section with* $\Phi(0) = 1$ *is*

$$\Phi(t) = \frac{\sqrt{Q_L(t)}}{2\kappa},$$

and $H(t) = 2\kappa\, t^2\, \Phi(t)$.

Proof. Part (i): $Q_L'/(2Q_L)\,dt$ is closed on a one-dimensional base. Parts (ii)–(iii): standard properties of the Gauss–Manin connection of $\{\sqrt{Q_L(t)}\}_t$. Near a simple zero t_0 of Q_L: $Q_L(t) \approx Q_L'(t_0)(t-t_0)$, so $Q_L'/(2Q_L)\,dt \approx dt/(2(t-t_0))$. The monodromy of a square root around a simple branch point is -1. Part (iv): direct verification. □

9.2. **Koszul duality and discriminant complementarity.**

Proposition 9.3 (Discriminant complementarity)**.** *For the Virasoro family (*$\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$*):*

$$\Delta(\mathrm{Vir}_c) + \Delta(\mathrm{Vir}_{26-c}) \;=\; \frac{6960}{(5c+22)(152-5c)}.$$

At the self-dual point $c = 13$*:* $\Delta(13) = 40/87$.

Proof.

$$\begin{aligned}\Delta(c) + \Delta(26-c) \;&=\; \frac{40}{5c+22} + \frac{40}{152-5c}\\ &=\; 40\cdot\frac{174}{(5c+22)(152-5c)} \;=\; \frac{6960}{(5c+22)(152-5c)}.\end{aligned}$$ □

Remark 9.4 (The self-dual point)*.* At $c = 13$, the Koszul involution $\mathrm{Vir}_c \mapsto \mathrm{Vir}_{26-c}$ has a fixed point: Vir_{13} is self-dual under the uncurved quadratic Koszul duality $c \mapsto 26-c$. The shadow metric satisfies $Q_L = Q_L^!$, the connection is self-dual, and the shadow growth rates are equal: $\rho = \rho^!$. The Koszul scalar complementarity $\kappa + \kappa' = 13$ (Virasoro-specific; not universal) reflects this fixed-point structure. The scalar quantity $\kappa + \kappa^!$ is distinct from the Trinity ghost-charge conductor $K(\mathrm{Vir}) = c + c^! = 26$.

9.3. **The Verdier involution on shadow data.** Under Koszul duality $\mathcal{A} \mapsto \mathcal{A}^!$, the shadow data transforms. The transformation depends on the family.

Proposition 9.5 (Koszul transformation of shadow data)**.**

(i) Affine Kac–Moody. $V_k(\mathfrak{g})^! = \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_{-k-2h^\vee})$. *The shadow data transforms as* $\kappa \mapsto -\kappa$*,* $\alpha \mapsto -\alpha$*,* $S_4 \mapsto 0$*. The Koszul conductor is* $\kappa(V_k(\mathfrak{g})) + \kappa(V_k(\mathfrak{g})^!) = 0$.

(ii) Virasoro. $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$. *The shadow data transforms by* $c \mapsto 26-c$*:* $\kappa \mapsto (26-c)/2$*,* $\alpha \mapsto 2$ *(invariant),* $S_4 \mapsto 10/[(26-c)(152-5c)]$*. The Koszul conductor is* $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = 13$.

(iii) Heisenberg. $\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ *(the chiral symmetric algebra on the dual). Note:* $\mathcal{H}_k^! \neq \mathcal{H}_{-k}$ *(these are isomorphic as vertex algebras but not as Koszul duals; the Koszul dual has the same* κ*). The Koszul conductor is* $\kappa + \kappa' = 0$.

Remark 9.6 (Family-dependence of the scalar complementarity)*.* The scalar complementarity invariant $\varkappa(\mathcal{A}) := \kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$ is *not* universal: it depends on the family. For affine Kac–Moody and free fields: $\varkappa = 0$. For Virasoro: $\varkappa = 13$. For $\mathcal{W}_3$: $\varkappa = 250/3$. For Bershadsky–Polyakov: $\varkappa = 98/3$. The statement "$\kappa + \kappa' = 0$" without a family qualifier is false for Virasoro and all $\mathcal{W}_N$ with $N \geq 2$. The scalar quantity $\varkappa = \kappa + \kappa^!$ is distinct from the Trinity ghost-charge conductor $K(\mathcal{A}) := c(\mathcal{A}) + c(\mathcal{A}^!) = -c_{\mathrm{ghost}}(\mathrm{BRST})$, which takes per-family values $K(\mathrm{Vir}) = 26$, $K(\mathcal{W}_3) = 100$, $K(\mathrm{BP}) = 196$, $K(V_k(\mathfrak{g})) = 2\dim\mathfrak{g}$, $K(\mathcal{W}_N) = 4N^3 - 2N - 2$. The two are related by $\varkappa = \varrho_{\mathcal{A}}\cdot K$ with $\varrho_{\mathcal{W}_N} = H_N - 1$, $\varrho_{\mathrm{KM}} = \varrho_{\mathrm{free}} = 0$, $\varrho_{\mathrm{BP}} = 1/6$.

9.4. **The shadow connection for each class.** The shadow connection $\nabla^{\mathrm{sh}} = d - Q_L'/(2Q_L)\,dt$ has qualitatively different behaviour in each class.

Example 9.7 (Class **G**: trivial)**.** $Q_L = 4\kappa^2$, constant. $Q_L' = 0$, so $\nabla^{\mathrm{sh}} = d$: the connection is trivial, the flat section is $\Phi(t) = 1$, and there are no branch points. The shadow tower carries no monodromy.

Example 9.8 (Class **L**: single branch point). $Q_L = (2\kappa + 3\alpha t)^2$, a perfect square with a double zero at $t_0 = -2\kappa/(3\alpha)$. The connection form $Q_L'/(2Q_L)\,dt = 3\alpha/(2\kappa + 3\alpha t)\,dt$ has a simple pole with residue 1 at t_0. The monodromy is $\exp(2\pi i \cdot 1) = 1$: trivial.

The double zero of Q_L means that $\sqrt{Q_L}$ is single-valued (polynomial), consistent with the termination of the tower at degree 3.

Example 9.9 (Class **M**: Virasoro). $Q_L = (c + 6t)^2 + 80t^2/(5c + 22)$. The two simple zeros

$$t_\pm = \frac{-3c \pm ic\sqrt{40/((5c+22)c^2 - 36)}}{9 + 40/(5c+22)}$$

form a complex conjugate pair (for $c > 0$ real). Each has residue 1/2; monodromy -1. The shadow connection is the Gauss–Manin connection of the double cover $\Sigma_L = \{y^2 = Q_L(t)\}$, and the branch-point monodromy is the deck transformation $y \mapsto -y$.

9.5. The planted-forest formula.

The MC equation at genus 2 produces a codimension-2 boundary correction.

Proposition 9.10 (Planted-forest correction;). *In the signed planted-forest chain model, the predicted genus-2 planted-forest correction on a primary line is*

$$\delta_{\mathrm{pf}}^{(2,0)} = \frac{S_3(10\,S_3 - \kappa)}{48},$$

provided the determinant-line signs make the planted-forest boundary operator square to zero and the residue-pushforward transport from the codimension-2 logarithmic Fulton–MacPherson boundary has been fixed. With that orientation package in place, the expression measures the contribution of codimension-2 boundary strata to the genus-2 amplitude. This correction vanishes for class **G** *(where $S_3 = 0$). For Virasoro: $\delta_{\mathrm{pf}}^{(2,0)} = -(c - 40)/48$.*

Proof. The codimension-2 boundary strata of $\overline{\mathcal{M}}_{2,0}$ correspond to stable graphs with two internal edges. After choosing the determinant-line orientation for the two possible contraction orders, each edge carries the propagator $P = 1/\kappa$ and connects two cubic vertices contributing S_3. The planted-forest graphs (two disjoint trees planted on a genus-1 surface) contribute a quadratic expression in S_3. The linear correction $-\kappa\,S_3/48$ comes from the interaction between the planted tree and the genus-1 curvature κ. The displayed formula is the resulting chain-level computation; without the signed orientation and residue-pushforward package it is a conditional model statement, not a standalone geometric consequence of the logarithmic compactification. □

10. Explicit shadow towers for the standard landscape

We compute the shadow data $(\kappa, \alpha, S_4, \Delta, \rho)$ for all standard families, constructing each tower from the Heisenberg base case upward.

10.1. Class G: Heisenberg.

Proposition 10.1 (Heisenberg shadow tower). *For the rank-1 Heisenberg vertex algebra $\mathcal{H}_k$ at level $k \neq 0$:*

$$\kappa = k, \qquad \alpha = 0, \qquad S_r = 0 \quad \textit{for all } r \geq 3.$$

The tower terminates at degree 2: $Q_L(t) = 4k^2$ (constant), $H(t) = 2k\,t^2$. Class **G**, *$d_{\mathrm{alg}} = 0$. The genus-g free energy is $F_g(\mathcal{H}_k) = k \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT) for all $g \geq 1$.*

Proof. The Heisenberg OPE $J(z)J(w) \sim k/(z-w)^2$ is purely quadratic: all collision residues on $\mathrm{FM}_n(X)$ for $n \geq 3$ vanish. Hence $\alpha = 0$, $S_4 = 0$, $\Delta = 0$. The shadow metric is the constant $4k^2$, a perfect square. □

Proposition 10.2 (Lattice VOA shadow tower). *For the lattice vertex algebra V_Λ associated to an even positive-definite lattice Λ of rank r:*

$$\kappa = r, \qquad \alpha = 0, \qquad S_n = 0 \quad \text{for all } n \geq 3.$$

Class **G**, *independent of the lattice cocycle.*

Proof. The primary line of a lattice VOA is spanned by the Heisenberg currents J^i, $i = 1, \ldots, r$. The OPE $J^i(z)J^j(w) \sim \delta^{ij}/(z-w)^2$ is abelian: no cubic or higher terms on the primary line. The lattice vertex operators e^α have nontrivial OPEs but live on charged strata; they do not contribute to the primary-line shadow tower. The modular characteristic $\kappa = r$ equals the rank, independent of the specific lattice structure (self-dual, Niemeier, Leech, etc.). □

Remark 10.3 (Free fermion). The free fermion ψ with $\psi(z)\psi(w) \sim 1/(z-w)$ has $c = 1/2$ and $\kappa = c/2 = 1/4$. On the single-generator primary line, the OPE is purely quadratic (Clifford), so $\alpha = 0$, $S_4 = 0$, $\Delta = 0$. Class **G**, $r_{\max} = 2$, $d_{\mathrm{alg}} = 0$.

Remark 10.4 (Class **G** as the base case). Class **G** is the base case of the shadow obstruction tower: the tower terminates at the curvature κ, and all higher invariants vanish. This is the vertex-algebraic analogue of the classical Koszul property: the bar resolution is minimal, the A_∞ structure is formal (all $m_k = 0$ for $k \geq 3$), and the genus-g free energies are determined by a single invariant. The entire moduli-space content of a class **G** algebra is encoded in the number κ.

The transition from class **G** to class **L** is the introduction of a Lie bracket: the cubic shadow $\alpha \neq 0$ measures the failure of the OPE to be abelian. The transition from class **L** to class **M** is the introduction of a nonlinear contact term: the quartic shadow $S_4 \neq 0$ measures the failure of the Jacobi identity to kill the four-point function. Each transition introduces new invariants and new complexity.

10.2. Class L: affine Kac–Moody.

Proposition 10.5 (Affine Kac–Moody shadow tower). *For $V_k(\mathfrak{g})$ at level $k \neq -h^\vee$ with $\mathfrak{g}$ simple:*

$$\kappa = \frac{\dim(\mathfrak{g})\,(k+h^\vee)}{2h^\vee}, \qquad \alpha \neq 0, \qquad S_4 = 0, \qquad \Delta = 0.$$

The cubic shadow is nonzero (generated by the Lie bracket), and the quartic shadow vanishes on the primary line (killed by the Jacobi identity). The tower terminates at degree 3: $r_{\max} = 3$, class **L**, *$d_{\mathrm{alg}} = 1$.*

Boundary values: at $k = 0$ (abelian limit), $\kappa = \dim(\mathfrak{g})/2 \neq 0$ (the algebra is still Koszul); at $k = -h^\vee$ (critical level), $\kappa = 0$ (Koszulness fails, the center jumps).

Proof. The modular characteristic $\kappa = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$ is the genus-1 curvature scalar. The derivation: the OPE $J^a(z)J^b(w) \sim k\,\delta^{ab}/(z-w)^2 + f^{abc}J^c(w)/(z-w)$ has a double pole $k\delta^{ab}$ and a simple pole $f^{abc}J^c$. The R-matrix in trace-form convention is $r(z) = k\Omega/z$ (with level prefix k; at $k = 0$, $r(z) = 0$). Averaging: $\mathrm{av}(k\Omega/z) = k\,\dim(\mathfrak{g})/(2h^\vee)$. The Sugawara shift $\dim(\mathfrak{g})/2$ arises from the simple-pole self-contraction through the adjoint Casimir eigenvalue $2h^\vee$. Total: $\kappa = k\,\dim(\mathfrak{g})/(2h^\vee) + \dim(\mathfrak{g})/2 = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$.

The cubic shadow α encodes the Lie bracket: the three-point collision residue on $\mathrm{FM}_3(X)$ extracts f^{abc} from the simple-pole OPE. On the primary line, α is a nonzero scalar proportional to the cubic Casimir.

The quartic collision residue on $\mathrm{FM}_4(X)$, restricted to the primary line, is proportional to the cyclic sum $\sum_{\mathrm{cyclic}(a,b,c)} f^{abx} f^{xcd}$, which vanishes by the Jacobi identity $[[\cdot,\cdot],\cdot] + \text{cyclic} = 0$. Hence $S_4 = 0$, $\Delta = 8\kappa \cdot 0 = 0$, and $Q_L = (2\kappa + 3\alpha\, t)^2$.

Verification at boundary values: at $k = 0$ (abelian limit), $\kappa = \dim(\mathfrak{g})/2 \neq 0$ for non-abelian $\mathfrak{g}$. The R-matrix vanishes but the Sugawara shift persists; the algebra remains Koszul. At $k = -h^\vee$ (critical level), $\kappa = 0$: the modular characteristic vanishes, the center of the completed enveloping algebra jumps (from $\mathbb{C}$ to $\mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}})$), and Koszulness fails. □

Remark 10.6 (The Jacobi identity as depth controller). The vanishing $S_4 = 0$ for affine Kac–Moody is not a numerical coincidence: it is a structural consequence of the Jacobi identity. The class **L** designation reflects the Lie-theoretic origin of the algebra: the tower terminates because the Lie bracket closes (Jacobi), not because of any numerical cancellation in the OPE coefficients. This explains why the transition from class **L** to class **M** occurs precisely when the Jacobi identity is replaced by a more general relation (the Virasoro commutator, the $\mathcal{W}_N$ OPE).

The modular characteristics for small Lie algebras at $k = 1$:

$\mathfrak{g}$	$\dim(\mathfrak{g})$	$h^\vee$	$\kappa(k)$	$\kappa(1)$
$\mathfrak{sl}_2$	3	2	$3(k+2)/4$	$9/4$
$\mathfrak{sl}_3$	8	3	$4(k+3)/3$	$16/3$
$\mathfrak{sl}_N$	$N^2{-}1$	N	$(N^2{-}1)(k{+}N)/(2N)$	--
E_8	248	30	$248(k{+}30)/60$	$1922/15$

The Koszul dual $V_k(\mathfrak{g})^! = \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_{-k-2h^\vee})$ has $\kappa^! = -\kappa$. The Koszul conductor is family-specific: $\kappa(V_k(\mathfrak{g})) + \kappa(V_k(\mathfrak{g})^!) = 0$ for affine Kac–Moody and free fields; $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = 13$ for Virasoro. These are distinct: the vanishing for Kac–Moody reflects the anti-symmetry of the chiral Chevalley–Eilenberg construction.

10.3. Class C: $\beta\gamma$.

Proposition 10.7 ($\beta\gamma$ shadow tower). *For the $\beta\gamma$ system of conformal weight λ:*

$$\begin{aligned} \kappa &= 6\lambda^2 - 6\lambda + 1 = c/2, & c &= 2(6\lambda^2 - 6\lambda + 1),\\ \alpha &= 0 & &\text{(weight-changing line)},\\ S_4 &= 0 & &\text{(weight-changing line)},\\ \Delta &= 0 & &\text{(weight-changing line)}. \end{aligned}$$

On the weight-changing primary line, the tower terminates at degree 2. The overall classification is class **C***: the standard conformal-weight family has $S_3 = 0$, $S_4 = -5/12$, and $S_r = 0$ for $r \geq 5$. Hence $r_{\max} = 4$, $d_{\mathrm{alg}} = 2$.*

Boundary values: $\lambda = 1/2$ gives $c = -1$ (symplectic boson), $\kappa = -1/2$; $\lambda = 2$ gives $c = 26$ (matter ghost), $\kappa = 13$.

Proof. The $\beta\gamma$ system has fields β of weight λ and γ of weight $1 - \lambda$, with OPE $\beta(z)\gamma(w) \sim 1/(z - w)$. The central charge is $c_{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1)$, and $\kappa = c/2 = 6\lambda^2 - 6\lambda + 1$.

On the weight-changing primary line (spanned by $\beta\gamma$ composites of definite weight), the OPE is abelian: the Wick contraction of β with γ produces only the identity, with no higher poles. Hence $\alpha = 0$ and $S_4 = 0$ on this line, and $\Delta = 0$.

The standard conformal-weight family formulas are $S_3 = 0$, $Q^{\mathrm{cont}} = S_4 = -5/12$, and $S_r = 0$ for $r \geq 5$. Thus the full family is class **C**, whereas the weight-changing primary line by itself is shadow-trivial. Hence the tower terminates at degree 4.

Boundary values: $c_{\beta\gamma}(1/2) = -1$ (symplectic boson), $\kappa = -1/2$; $c_{\beta\gamma}(2) = 26$ (matter ghost), $\kappa = 13$; $c_{\beta\gamma}(0) = 2$ (standard $\beta\gamma$), $\kappa = 1$; $c_{\beta\gamma}(1) = 2$ (adjoint $\beta\gamma$), $\kappa = 1$. □

Remark 10.8 (bc ghosts). The fermionic bc system has $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$ and $\kappa_{bc} = c_{bc}/2 = -(6\lambda^2 - 6\lambda + 1)$. The complementarity $c_{\beta\gamma}(\lambda) + c_{bc}(\lambda) = 0$ holds at each conformal weight: $2(6\lambda^2 - 6\lambda + 1) + 1 - 3(2\lambda - 1)^2 = 0$ (direct expansion). At $\lambda = 2$ (reparametrization ghosts): $c_{bc} = -26$, $\kappa_{bc} = -13$. At $\lambda = 1/2$ (single Dirac fermion): $c_{bc} = 1$, $\kappa_{bc} = 1/2$. Class **C** for all λ.

Remark 10.9 (The contact stratum geometry). Class **C** is the unique class where the shadow tower terminates at a finite depth greater than 3. The mechanism is geometric: the quartic contact invariant lives on a stratum of the cyclic deformation complex that is *transverse* to every one-dimensional primary line. On any primary line, the shadow data is class **G** (no cubic, no quartic); the class **C** designation comes from the global structure of the deformation complex.

This transversality is specific to the $\beta\gamma/bc$ family: the weight-changing direction (which supports the primary-line tower) is orthogonal to the ghost-number direction (which supports the quartic contact). For Virasoro and $\mathcal{W}_N$, the quartic contribution lies on the primary line itself, producing $S_4 \neq 0$ and $\Delta \neq 0$ (class **M**). For Heisenberg and affine KM, the quartic contribution vanishes entirely (abelian or Jacobi).

Example 10.10 ($\beta\gamma$ at special weights).

λ	$c_{\beta\gamma}$	κ	Interpretation	c_{bc}
0	2	1	standard	−2
1/2	−1	−1/2	symplectic boson	1
1	2	1	adjoint	−2
3/2	10	5	gravitino	−10
2	26	13	matter ghost	−26

At $\lambda = 2$, the matter ghost has $c = 26$, which is the string-theoretic value: the $\beta\gamma$ matter ghost cancels the bc reparametrization ghost central charge $(26 + (-26) = 0)$. The modular characteristics also cancel: $13 + (-13) = 0$.

10.4. Class M: Virasoro.

Proposition 10.11 (Virasoro shadow tower). *For* Vir_c *at* $c \neq 0$, $c \neq -22/5$*:*

$$\kappa = \frac{c}{2}, \qquad \alpha = 2, \qquad S_4 = \frac{10}{c(5c+22)}, \qquad \Delta = \frac{40}{5c+22}.$$

The shadow metric is

$$Q_L(t) = (c+6t)^2 + \frac{80\,t^2}{5c+22}.$$

The tower is infinite (class **M***,* $d_{\mathrm{alg}} = \infty$*), and the shadow growth rate is* $\rho = \frac{1}{c}\sqrt{(180c+872)/(5c+22)}$*.*

Remark 10.12 (The quartic contact invariant). The quartic shadow $Q^{\mathrm{contact}}_{\mathrm{Vir}} = S_4 = 10/[c(5c+22)]$ is the first genuinely nonlinear OPE invariant. It measures the irreducible four-point collision residue on $\mathrm{FM}_4(X)$ and is the coefficient that separates class **M** from class **L**. Its poles at $c = 0$ and $c = -22/5$ reflect the failure of the Virasoro OPE at these special values. The large-c asymptotic is $S_4 \sim 2/c^2$.

Theorem 10.13 (Virasoro shadow coefficients)**.** *The shadow coefficients through degree* 10*:*

$$\begin{aligned}
S_2 &= c/2,\\
S_3 &= 2,\\
S_4 &= \frac{10}{c(5c+22)},\\
S_5 &= \frac{-48}{c^2(5c+22)},\\
S_6 &= \frac{80(45c+193)}{3\,c^3(5c+22)^2},\\
S_7 &= \frac{-2880(15c+61)}{7\,c^4(5c+22)^2},\\
S_8 &= \frac{80(2025c^2+16470c+33314)}{c^5(5c+22)^3},\\
S_9 &= \frac{-1280(2025c^2+15570c+29554)}{3\,c^6(5c+22)^3},\\
S_{10} &= \frac{256(91125c^3+1050975c^2+3989790c+4969967)}{c^7(5c+22)^4}.
\end{aligned}$$

All coefficients belong to $\mathbb{Q}(c)$ *with poles only at* $c = 0$ *and* $c = -22/5$*. Signs alternate from* S_5 *onward:* $S_5 < 0$, $S_6 > 0$, $S_7 < 0, \ldots$ *for* $c > 0$*. The* c*-independence of* $S_3 = 2$ *is a theorem; see* [18].

Proof. The computation proceeds inductively using the master equation (Proposition 3.2) on the primary line L spanned by T. The Hessian propagator is $P = 1/\kappa = 2/c$.

Degree 2. $S_2 = \kappa = c/2$ by definition: this is the genus-1 curvature scalar $F_1 = \kappa/24 = c/48$.

Degree 3. The OPE $T_{(1)}T = 2T$ gives the simple-pole collision residue (after $d\log$ absorption from the $T_{(2)}T = \partial T$ term). On the primary line: $S_3 = \alpha = 2$, independent of c. This c-independence follows from the universality of the Virasoro commutation relation $[L_m, L_n] = (m-n)L_{m+n} + (c/12)(m^3 - m)\delta_{m+n,0}$: the coefficient of L_{m+n} is $m-n$, not c-dependent. The scalar $\alpha = 2$ is twice the coefficient of T in $T_{(1)}T = 2T$.

Degree 4. The master equation at degree 4 is $d_0(\mathrm{Sh}_4) + \frac{1}{2}[\mathrm{Sh}_3, \mathrm{Sh}_3] = 0$. On the primary line: $8\kappa S_4 + \frac{1}{2}\,c_{33}\,9\,P\,\alpha^2 = 0$, where $c_{33} = 1/2$ (self-pairing correction). Solving:

$$S_4 = -\frac{1}{8\kappa}\cdot\frac{1}{2}\cdot 9\cdot P\cdot\alpha^2 = -\frac{9\alpha^2}{16\kappa^2} + S_4^{\mathrm{contact}},$$

where S_4^{contact} is the irreducible quartic contact term from the four-point OPE. For Virasoro, the quartic OPE coefficient is $T_{(3)}T = c/2$, and the contact extraction gives $S_4^{\mathrm{contact}} = 10/[c(5c+22)]$ after the propagator renormalization.

The full result: $S_4 = 10/[c(5c+22)]$, and the critical discriminant is $\Delta = 8\kappa S_4 = 8(c/2)\cdot 10/[c(5c+22)] = 40/(5c+22) \neq 0$ for $c \neq -22/5$.

Degree 5. $d_0(\mathrm{Sh}_5) + [\mathrm{Sh}_3, \mathrm{Sh}_4] = 0$, giving $S_5 = -(P/10)\cdot 3\cdot 4\cdot S_3S_4 = -(6/5)(2/c)\cdot 2\cdot 10/[c(5c+22)] = -48/[c^2(5c+22)]$.

Higher degrees. The pattern continues: each S_r involves a sum over pairs (j,k) with $j+k = r+2$, weighted by the propagator $P = 2/c$ and the symmetry factors c_{jk}. Alternatively, the closed-form extraction $S_r = \frac{1}{r}[t^{r-2}]\sqrt{Q_{\mathrm{Vir}}(t)}$ with $Q_{\mathrm{Vir}}(t) = (c+6t)^2 + 80t^2/(5c+22)$ gives all coefficients by binomial expansion.

Both methods have been verified computationally through degree 20; the results agree to all digits. □

Remark 10.14 (Pole structure of Virasoro shadows). Every S_r with $r \geq 4$ has poles only at $c = 0$ and $c = -22/5$. The pole at $c = 0$ has order $r - 3$; the pole at $c = -22/5$ has order $\lfloor (r-2)/2 \rfloor$. This pole structure is inherited from the branch points of $\sqrt{Q_{\mathrm{Vir}}(t)}$: the zero of Q_{Vir} nearest to $t = 0$ has modulus inversely proportional to the shadow growth rate ρ, and the Taylor coefficients of a square-root function at a simple branch point have a universal pole pattern determined by the Flajolet–Sedgewick transfer theorem.

The physical content: the pole at $c = 0$ reflects the vanishing of the Virasoro central charge (the algebra degenerates to the Witt algebra, which has no central extension and no modular characteristic). The pole at $c = -22/5$ is the second zero of the discriminant of the Virasoro Verma module at level 2: at this value, the four-point contact term diverges, and the shadow metric Q_{Vir} degenerates.

Numerical evaluations:

r	$S_r(c{=}1)$	$S_r(c{=}13)$	$S_r(c{=}26)$
2	1/2	13/2	13
3	2	2	2
4	10/27	10/1131	5/1976
5	−16/9	−16/4901	−3/6422

At $c = 26$ the coefficients decay rapidly ($\rho \approx 0.232$); at $c = 13$ moderately ($\rho \approx 0.467$); at $c = 1$ they grow exponentially ($\rho \approx 6.242$). All unitary minimal models ($c < 1$) have strongly divergent towers.

Extended numerical data through degree 8:

r	$S_r(c{=}1)$	$S_r(c{=}13)$	$S_r(c{=}26)$
6	19040/2187	62240/49887279	6815/76139232
7	−24320/567	−81920/168138607	−20295/1154778352
8	4144720/19683	47171920/244497554379	4576085/1303909727744

The alternating signs $(-1)^{r+1}$ for $r \geq 5$ (at $c > 0$) are a consequence of the complex conjugate branch points of Q_L: the transfer theorem gives oscillating coefficients with frequency determined by the argument of the nearest singularity.

Remark 10.15 (The Virasoro shadow at special values). Three special values of c merit attention:

$c = 13$ *(self-dual).* Under Koszul duality $c \mapsto 26 - c$, the self-dual point $c = 13$ maps to itself. The shadow data is symmetric: $\kappa = 13/2$, $\kappa' = 13/2$, $\alpha = \alpha' = 2$, $S_4(13) = S_4(13)$. The discriminant $\Delta(13) = 40/87$, and the shadow growth rate $\rho(13) \approx 0.467$. All shadow coefficients are self-dual: $S_r(\mathrm{Vir}_{13}) = S_r(\mathrm{Vir}^!_{13})$.

$c = 26$ *(string).* The Koszul dual Vir_0 has $\kappa = 0$ (the bar complex is uncurved). The shadow tower of Vir_{26} is well-behaved ($\rho \approx 0.232$), but the dual tower is degenerate. The discriminant $\Delta(26) = 40/152 = 5/19$.

$c = -22/5$ *(pole).* This is the location of the denominator singularity $5c + 22 = 0$. The quartic shadow S_4 diverges, $\Delta \to \infty$, and the shadow metric degenerates. This value is associated with the $(2, 5)$ minimal model in the non-unitary series. The shadow tower is undefined at $c = -22/5$ but has well-defined limits from both sides.

Proposition 10.16 (The Virasoro shadow connection: explicit formula). *The shadow connection for Virasoro is*

$$\nabla^{\mathrm{sh}} \;=\; d \;-\; \frac{6c + (18 + 80/(5c+22))\,t}{(c+6t)^2 + 80t^2/(5c+22)}\,dt.$$

The connection has two regular singular points at the zeros of $Q_{\mathrm{Vir}}(t)$*, each with residue* $1/2$*. The zeros are complex conjugate for* $c > 0$*:*

$$t_\pm \;=\; \frac{-3c(5c+22) \pm ic\sqrt{80(5c+22)}}{(9(5c+22)+80)}.$$

The branch-point modulus is $|t_\pm| = c/\sqrt{9 + 80/(5c+22)} = 2\kappa/\sqrt{9\alpha^2 + 2\Delta} = 1/\rho$*, confirming the shadow growth rate formula.*

Remark 10.17 (Duality and the shadow connection). Under $c \mapsto 26 - c$, the connection form transforms by a rational substitution:

$$\left.\frac{Q_L'(t)}{2Q_L(t)}\right|_c \;\neq\; \left.\frac{Q_L'(t)}{2Q_L(t)}\right|_{26-c}.$$

The two connections are related by the Verdier involution on the shadow data, but they are not isomorphic as connections on the same base: the branch points move under $c \mapsto 26 - c$. The discriminant complementarity $\Delta(c) + \Delta(26-c) = 6960/[(5c+22)(152-5c)]$ (Proposition 9.3) reflects this transformation.

10.5. **The shadow depth decomposition.** For class **M** algebras, the shadow tower is infinite. The complexity of the tower admits a decomposition into algebraic and arithmetic components.

Definition 10.18 (Shadow depth decomposition). For a chirally Koszul vertex algebra $\mathcal{A}$ of class **M**, define the *total shadow depth*

$$d(\mathcal{A}) \;=\; 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}},$$

where:

- $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ is the algebraic depth (Definition 6.1). For class **M**: $d_{\mathrm{alg}} = \infty$ (all S_r nonzero), but the algebraicity theorem shows that three invariants determine the full tower, so the "effective algebraic complexity" is 0.
- $d_{\mathrm{arith}} \geq 0$ is the *arithmetic depth*: the number of genuinely independent arithmetic invariants beyond the algebraic shadow data. Shadow coefficients of weight $\leq 2r$ at degree r lie in the subspace spanned by Eisenstein series; cusp-form contributions first appear at genus $g \geq 6$ (the Ramanujan Δ form at weight 12).

Remark 10.19 (Algebraic vs arithmetic invariants). The algebraic depth is determined by genus-0 OPE data (Theorem 4.1). The arithmetic depth reflects automorphic phenomena in the genus expansion: cusp-form contributions are invisible at low genus but may enter at $g \geq 6$. For all families treated in Section 10, $d_{\mathrm{arith}} = 0$ through the genera computed here.

The distinction between algebraic and arithmetic depth is the shadow-tower analogue of the distinction between perturbative and non-perturbative contributions in quantum field theory: the algebraic part is determined by finitely many OPE data, while the arithmetic part involves new transcendental numbers (cusp-form coefficients) that are not determined by the OPE.

10.6. **Shadow growth rate.**

Definition 10.20 (Shadow growth rate). For class **M** with $\Delta \neq 0$:

$$\rho(\mathcal{A}) \;:=\; \frac{\sqrt{9\alpha^2 + 2\Delta}}{2|\kappa|}.$$

Proposition 10.21. *The growth rate $\rho(\mathcal{A})$ is the reciprocal of the convergence radius of $H(t) = t^2\sqrt{Q_L(t)}$. The branch points of H are the zeros of Q_L. For $\Delta > 0$, these form a complex conjugate pair of modulus $R = 2|\kappa|/\sqrt{9\alpha^2 + 2\Delta}$, and*

$$S_r \;\sim\; C(\mathcal{A})\,\rho(\mathcal{A})^r\,r^{-5/2}\,\cos(r\theta + \phi), \qquad r \to \infty,$$

where the exponent $-5/2$ comes from the square-root singularity (Flajolet–Sedgewick transfer theorem).

Example 10.22 (Virasoro shadow growth rate)**.** For Vir_c with $c > 0$:

$$\rho(\mathrm{Vir}_c) \;=\; \frac{1}{c}\sqrt{\frac{180c + 872}{5c + 22}}.$$

The critical central charge $c^* \approx 6.125$ (the unique positive root of $5c^3 + 22c^2 - 180c - 872 = 0$) separates the convergent regime ($c > c^*$, $\rho < 1$) from the divergent regime ($c < c^*$, $\rho > 1$).

Koszul duality exchanges convergent and divergent towers:

c	$\rho(c)$	$\rho(26{-}c)$	converges?	interpretation
$1/2$	12.53	0.224	no/yes	Ising
1	6.242	0.242	no/yes	free boson
c^*	1	0.430	marginal/yes	critical
13	0.467	0.467	yes/yes	self-dual
26	0.232	∞	yes/–-	string; $\kappa^! = 0$

10.7. Class M: $\mathcal{W}_N$.

Proposition 10.23 ($\mathcal{W}_N$ shadow data)**.** *For $\mathcal{W}_N$ ($N \geq 3$) at central charge c, the total modular characteristic is*

$$\kappa(\mathcal{W}_N) \;=\; c\,(H_N - 1), \qquad H_N \;=\; \sum_{j=1}^{N} \frac{1}{j}.$$

On the T-line (Virasoro direction): $\kappa_T = c/2$, $\alpha_T = 2$, $S_4 = 10/[c(5c + 22)]$, class **M**.

The anomaly ratios for small N:

N	$\mathcal{W}_N$	$H_N - 1$	$\kappa(\mathcal{W}_N)$
2	Virasoro	$1/2$	$c/2$
3	$\mathcal{W}_3$	$5/6$	$5c/6$
4	$\mathcal{W}_4$	$13/12$	$13c/12$
5	$\mathcal{W}_5$	$77/60$	$77c/60$
6	$\mathcal{W}_6$	$29/20$	$29c/20$

Sanity check at $N = 2$: $H_2 - 1 = 3/2 - 1 = 1/2$, so $\kappa(\mathcal{W}_2) = c/2 = \kappa(\mathrm{Vir}_c)$. At $N = 3$: $H_3 - 1 = 11/6 - 1 = 5/6$, so $\kappa(\mathcal{W}_3) = 5c/6$.

10.8. Master table.

Family	κ	α	S_4	Δ	Class	$r_{\max}$	ρ
$\mathcal{H}_k$	k	0	0	0	**G**	2	0
V_Λ	$\mathrm{rk}(\Lambda)$	0	0	0	**G**	2	0
$V_k(\mathfrak{g})$	$\frac{\dim\mathfrak{g}\,(k+h^\vee)}{2h^\vee}$	$\neq 0$	0	0	**L**	3	0
$\beta\gamma_\lambda$	$6\lambda^2-6\lambda+1$	0^*	0^*	0^*	**C**	4	--
Vir_c	$c/2$	2	$\frac{10}{c(5c+22)}$	$\frac{40}{5c+22}$	**M**	∞	$\frac{1}{c}\sqrt{\frac{180c+872}{5c+22}}$
$\mathcal{W}_N$	$(H_N-1)\,c$	$\neq 0$	$\neq 0$	$\neq 0$	**M**	∞	--

10.9. **The discriminant across the landscape.** The critical discriminant $\Delta = 8\kappa S_4$ is the single number that separates finite towers from infinite ones. Its values across the standard landscape exhibit a sharp dichotomy:

Family	Δ	$\Delta = 0$?	Class
$\mathcal{H}_k$	0	yes	**G**
V_Λ	0	yes	**G**
Free fermion	0	yes	**G**
$V_k(\mathfrak{g})$	0	yes	**L**
$\beta\gamma_\lambda$	0^*	yes*	**C**
Vir_c	$40/(5c+22)$	no	**M**
$\mathcal{W}_3$	$\neq 0$	no	**M**
$\mathcal{W}_N$ $(N \geq 3)$	$\neq 0$	no	**M**

The vanishing $\Delta = 0$ for classes **G**, **L**, and **C** has three distinct mechanisms:

- Class **G**: the OPE is abelian ($S_4 = 0$ because there is no four-point connected correlator).
- Class **L**: the OPE is non-abelian but Lie ($S_4 = 0$ because the Jacobi identity kills the quartic collision residue).
- Class **C**: $S_4 = 0$ on every primary line, but the quartic contact invariant lives on a transverse stratum.

Proposition 10.24 (The discriminant as a Koszul invariant)**.** *The critical discriminant Δ is a Koszul invariant: it transforms covariantly under Koszul duality $\mathcal{A} \mapsto \mathcal{A}^!$. For Virasoro:*

$$\Delta(\mathrm{Vir}_c) = \frac{40}{5c+22}, \qquad \Delta(\mathrm{Vir}_{26-c}) = \frac{40}{152-5c}.$$

The discriminant never vanishes for $c \neq -22/5$; the Koszul dual never changes class.

Proof. Direct substitution: $\Delta(c) = 8 \cdot (c/2) \cdot 10/[c(5c+22)] = 40/(5c+22)$ and $\Delta(26-c) = 40/(5(26-c)+22) = 40/(152-5c)$. Both are nonzero for $c \neq -22/5$ and $c \neq 152/5 = 30.4$, respectively. The dual singularities $c = -22/5$ and $c = 152/5$ sum to 26: they are Koszul partners. □

Remark 10.25 (The discriminant locus). In the three-dimensional parameter space (κ, α, S_4), the discriminant locus $\Delta = 0$ is the hyperplane $\kappa S_4 = 0$, which decomposes as $\kappa = 0$ (degenerate: the propagator is undefined) and $S_4 = 0$ (the Jacobi locus). The complement $\Delta \neq 0$ is a dense open subset.

Every standard family lies on the discriminant locus except for Virasoro and $\mathcal{W}_N$ ($N \geq 3$). The $\beta\gamma$ family supplies the standard conformal-weight class-**C** witness. The classification is:

- $\kappa = 0$: critical (not chirally Koszul in standard sense).
- $S_4 = 0$, $\alpha = 0$: class **G**.
- $S_4 = 0$, $\alpha \neq 0$: class **L**.
- $S_3 = 0$, $S_4 \neq 0$, $S_r = 0$ for $r \geq 5$: class **C**.
- $S_4 \neq 0$ (on primary line): class **M**.

This recovers the four-class partition from the discriminant alone.

10.10. **Complementarity across families.** The Koszul conductor $K(\mathcal{A}) = \kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$ is family-specific. The standard landscape yields four distinct values:

Family	Koszul conductor K	$K = 0$?
$\mathcal{H}_k$	0	yes
$V_k(\mathfrak{g})$	0	yes
$\beta\gamma_\lambda$	0	yes
Lattice	0	yes
Vir_c	13	no
$\mathcal{W}_3$	250/3	no
Bershadsky–Polyakov	98/3	no

The vanishing $K = 0$ for abelian algebras and affine Kac–Moody is the anti-symmetry property of the chiral Chevalley–Eilenberg construction: the dual has $\kappa^! = -\kappa$. For Virasoro, the scalar complementarity $\varkappa = 13$ is the self-dual value: $\kappa(c) + \kappa(26 - c) = c/2 + (26 - c)/2 = 13$.

Remark 10.26 (The scalar complementarity is not the Trinity conductor). The scalar complementarity $\varkappa = \kappa + \kappa'$ is an invariant of the Koszul *pair* $(\mathcal{A}, \mathcal{A}^!)$. It is distinct from the Trinity ghost-charge conductor $K(\mathcal{A}) := c(\mathcal{A}) + c(\mathcal{A}^!) = -c_{\text{ghost}}(\text{BRST})$ (values $K(\text{Vir}) = 26$, $K(\mathcal{W}_3) = 100$, $K(\text{BP}) = 196$, $K(V_k(\mathfrak{g})) = 2\dim\mathfrak{g}$), and distinct again from the central charge c (an invariant of the algebra alone). The three quantities are related by $\varkappa = \varrho_{\mathcal{A}} \cdot K$ and $K = c + c^!$.

11. Genus-g free energies from the shadow tower

The shadow obstruction tower produces invariants of moduli spaces through graph sums over stable curves. The genus-g *free energy* $F_g(\mathcal{A})$ is the integral of the tautological class produced by the MC equation on $\overline{\mathcal{M}}_g$.

11.1. The single-generator formula.

Theorem 11.1 (Single-generator genus-g universality)**.** *For any chirally Koszul vertex algebra $\mathcal{A}$ with a single strong generator of conformal weight h (UNIFORM-WEIGHT):*

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} \tag{11.1}$$

for all $g \geq 1$, where $\lambda_g^{\mathrm{FP}} = \int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g$ is the Faber–Pandharipande integral.

Proof sketch. The genus-g free energy is a sum over stable graphs Γ of genus g with no marked points:

$$F_g(\mathcal{A}) \;=\; \sum_{\Gamma} \frac{1}{|\operatorname{Aut}(\Gamma)|} \prod_{e \in E(\Gamma)} P \prod_{v \in V(\Gamma)} V_{g_v, n_v},$$

where $P = 1/\kappa$ is the propagator and V_{g_v,n_v} is the vertex factor at a vertex of genus g_v with n_v half-edges. For a single-generator algebra, every edge carries the same channel. The vertex factors are determined by the shadow coefficients: $V_{0,3} = S_3 = \alpha$, $V_{0,4} = S_4$, and $V_{g,n}$ for $g \geq 1$ encodes the genus-g contribution.

The key observation is that for single-generator algebras, the graph sum telescopes: the combinatorics of automorphisms, vertex factors, and propagators combine to produce $\kappa \cdot \lambda_g^{\mathrm{FP}}$, where the Faber–Pandharipande integral absorbs all the graph-theoretic data. The telescoping uses the dilaton equation and the string equation on $\overline{\mathcal{M}}_{g,n}$ to reduce the sum to a single integral.

At genus 1: $F_1 = \kappa/24 = \kappa \cdot \lambda_1^{\mathrm{FP}}$ (unconditional, holds for all families including multi-generator). At genus 2: $F_2 = 7\kappa/5760 = \kappa \cdot \lambda_2^{\mathrm{FP}}$ (requires single-generator hypothesis). □

Remark 11.2 (The generating function). The Faber–Pandharipande coefficients are the Taylor coefficients of the $\hat{A}$-genus:

$$\sum_{g \geq 1} \lambda_g^{\mathrm{FP}} \quad {}^{2g} \;=\; \frac{\quad/2}{\sin(\quad/2)} - 1.$$

The first values are: $\lambda_1^{\mathrm{FP}} = 1/24$, $\lambda_2^{\mathrm{FP}} = 7/5760$, $\lambda_3^{\mathrm{FP}} = 31/967680$.

11.2. **The genus-2 verification.** The genus-2 formula $F_2 = 7\kappa/5760$ requires an explicit sum over the seven stable graphs. For a single-generator algebra with shadow data (κ, α, S_4):

Lemma 11.3 (Genus-2 single-generator sum). *The graph-by-graph contributions to F_2 for a single-generator chirally Koszul algebra are:*

Graph	*Contribution*	*Simplified*
Γ_0*: smooth*	$V_{2,0}$	$7\kappa/5760$
Γ_1*: figure-eight*	$P \cdot \kappa^2/(24^2 \cdot 2)$	$\kappa/1152$
Γ_2*: banana*	$P^2 S_4/8$	$S_4/(2\kappa)$
Γ_3*: dumbbell*	$P(\kappa/24)^2/2$	$\kappa/1152$
Γ_4*: theta*	$P^3\alpha^2/12$	$\alpha^2/(12\kappa^3)$
Γ_5*: tadpole*	$P^2\alpha(\kappa/24)/2$	$\alpha/(48\kappa)$
Γ_6*: barbell*	$P^3\alpha^2/8$	$\alpha^2/(8\kappa^3)$

The total $F_2 = \sum_\Gamma F_2(\Gamma)$ equals $7\kappa/5760$ after substitution of the shadow data and simplification. The verification uses the MC recursion at degree 4 to eliminate S_4 in favour of κ, α, and the quartic OPE data.

11.3. **Multi-generator failure at genus ≥ 2.** For multi-generator algebras, the single-generator formula (11.1) fails. The failure mechanism is the cross-channel correction: edges of stable graphs can carry different generators, producing mixed-channel amplitudes not captured by the scalar κ.

Proposition 11.4 (Multi-generator decomposition). *For a chirally Koszul vertex algebra with generators of distinct conformal weights (ALL-WEIGHT):*

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} \;+\; \delta F_g^{\mathrm{cross}}(\mathcal{A}), \tag{11.2}$$

where $\delta F_g^{\mathrm{cross}}$ is the cross-channel correction computed by summing over mixed-channel assignments on stable graphs. At genus 1: $\delta F_1^{\mathrm{cross}} = 0$ (unconditional). At genus ≥ 2: $\delta F_g^{\mathrm{cross}} \neq 0$ generically.

The $\mathcal{W}_3$ computation in Section 12 is the first explicit instance of (11.2): $\delta F_2^{\mathrm{cross}}(\mathcal{W}_3) = (c+204)/(16c) \neq 0$.

12. The $\mathcal{W}_3$ cross-channel computation at genus 2

12.1. **The $\mathcal{W}_3$ Frobenius data.** The $\mathcal{W}_3$ algebra has two strong generators: T (weight 2) and W (weight 3). The bar-complex CohFT has a two-dimensional Frobenius algebra:

Lemma 12.1 ($\mathcal{W}_3$ Frobenius data)**.**

(i) Curvature metric. $\eta_{TT} = \kappa_T = c/2$, $\eta_{WW} = \kappa_W = c/3$, $\eta_{TW} = 0$. *Total:* $\kappa(\mathcal{W}_3) = c/2 + c/3 = 5c/6$.

(ii) Propagators. $\eta^{TT} = 2/c$, $\eta^{WW} = 3/c$. *Both channels use the weight-*1 *bar propagator* $d\log E(z,w)$.

(iii) Structure constants. *The* $\mathbb{Z}_2$ *symmetry* $W \mapsto -W$ *kills all constants with an odd number of* W *indices:* $C_{TTT} = C_{TWW} = C_{WWT} = c$*;* $C_{TTW} = C_{WWW} = 0$.

Proof. Part (i): the curvature κ_i is the leading OPE pole coefficient after $d\log$ absorption. From $T_{(3)}T = c/2$: the quartic OPE pole $(c/2)/(z-w)^4$ becomes the cubic bar pole $(c/2)/z^3$ after $d\log$; the curvature $\kappa_T = c/2$. From $W_{(5)}W = c/3$: the sextic OPE pole $(c/3)/(z-w)^6$ becomes the quintic bar pole $(c/3)/z^5$ after $d\log$; the curvature $\kappa_W = c/3$. The cross-curvature $\eta_{TW} = 0$ because $T_{(3)}W = 0$ (no quartic pole in the TW OPE; $T_{(1)}W = 3W$ is the highest non-derivative pole).

Part (ii): the propagators are the inverse curvature matrix $\eta^{ij} = (\eta_{ij})^{-1}$. Since the metric is diagonal: $\eta^{TT} = 1/\eta_{TT} = 2/c$ and $\eta^{WW} = 1/\eta_{WW} = 3/c$. Both propagators carry the weight-1 bar measure $d\log E(z,w)$ (the prime form logarithmic derivative), regardless of the conformal weight of the generator: the bar propagator weight is always 1.

Part (iii): the three-point structure constants are extracted from the simple-pole OPE coefficients.

- $T_{(1)}T = 2T$: the structure constant $C^T_{TT} = 2$ (the coefficient of T in $T_{(1)}T$); lowering: $C_{TTT} = \eta_{TT} \cdot C^T_{TT} = (c/2)\cdot 2 = c$.
- $T_{(1)}W = 3W$: the structure constant $C^W_{TW} = 3$; lowering: $C_{TWW} = \eta_{WW} \cdot 3 = (c/3)\cdot 3 = c$.
- $W_{(3)}W = 2T$: the structure constant $C^T_{WW} = 2$; lowering: $C_{WWT} = \eta_{TT} \cdot 2 = (c/2)\cdot 2 = c$.

The universality $C_{TTT} = C_{TWW} = C_{WWT} = c$ is not a coincidence: it reflects the fact that all three constants are controlled by the Virasoro action (the stress tensor T generates conformal transformations acting on all fields).

The vanishing $C_{TTW} = 0$ and $C_{WWW} = 0$ follows from the $\mathbb{Z}_2$ automorphism $W \mapsto -W$: under this symmetry, $C_{TTW} \mapsto -C_{TTW}$ (one W-index), forcing $C_{TTW} = 0$; similarly $C_{WWW} \mapsto -C_{WWW}$ (three W-indices: odd), forcing $C_{WWW} = 0$. □

Remark 12.2 (Quartic vertex universality). The genus-0 four-valent vertex factor

$$V_{0,4}(i,i\mid j,j) = \sum_{m\in\{T,W\}} \eta^{mm}\, C_{iim}\, C_{jjm}$$

equals $2c$ for all channel pairs $(i,j) \in \{T,W\}^2$. Only the T-channel contributes: $\eta^{TT}C_{iiT}C_{jjT} = (2/c)\cdot c\cdot c = 2c$ (using $C_{TTT} = C_{WWT} = c$). The W-channel requires C_{iiW}, which is either $C_{TTW} = 0$ or $C_{WWW} = 0$ by the $\mathbb{Z}_2$ selection rule.

This universality simplifies the genus-2 graph sum: the banana graph (Γ_2) has a universal quartic vertex regardless of the channel assignment on its two edges. For $\mathcal{W}_N$ with $N \geq 4$, the additional generators (spins $4,\ldots,N$) break the $\mathbb{Z}_2$ symmetry, and the quartic vertex becomes channel-dependent.

Remark 12.3 (Feynman rules for the genus-2 graph sum). The amplitude of a genus-2 stable graph Γ with channel assignment $\sigma: E(\Gamma) \to \{T, W\}$ is

$$A(\Gamma, \sigma) = \frac{1}{|\mathrm{Aut}(\Gamma)|} \prod_{e \in E(\Gamma)} \eta^{\sigma(e), \sigma(e)} \prod_{v \in V(\Gamma)} V_v(g_v, \mathrm{channels}_v),$$

where the vertex factor V_v depends on the vertex type:

- Genus-0 trivalent vertex with half-edges (i, j, k): $V_v = C_{ijk}$.
- Genus-0 four-valent vertex with half-edges (i, i, j, j): $V_v = V_{0,4}(i, i \mid j, j) = 2c$.
- Genus-1 univalent vertex with half-edge i: $V_v = \kappa_i/24$ (the genus-1 one-point function).
- Genus-2 vertex with no half-edges: $V_v = F_2^{\mathrm{smooth}}$ (the smooth genus-2 contribution).

The cross-channel correction sums $A(\Gamma, \sigma)$ over all graphs Γ and all mixed channel assignments σ.

12.2. **Stable graphs at genus 2.** There are seven stable graphs of genus 2 with no marked points:

Graph	Structure	$\lvert\mathrm{Aut}\rvert$	h^1
Γ_0: smooth	$1 \times (g{=}2, \mathrm{val} = 0)$	1	0
Γ_1: figure-eight	$1 \times (g{=}1, \mathrm{val} = 2)$, 1 self-loop	2	1
Γ_2: banana	$1 \times (g{=}0, \mathrm{val} = 4)$, 2 self-loops	8	2
Γ_3: dumbbell	$(g{=}1, 1) + (g{=}1, 1)$, 1 bridge	2	0
Γ_4: theta	$(g{=}0, 3) + (g{=}0, 3)$, 3 bridges	12	2
Γ_5: tadpole	$(g{=}0, 3) + (g{=}1, 1)$	2	1
Γ_6: barbell	$(g{=}0, 3) + (g{=}0, 3)$, 2 loops + 1 bridge	8	2

Each edge carries a channel label $\sigma(e) \in \{T, W\}$. A channel assignment is *mixed* if not all edges carry the same channel. The cross-channel correction is the sum of mixed-assignment amplitudes.

12.3. **Graph-by-graph computation.** We compute the mixed-channel contribution from each graph.

Γ_0 *(smooth),* Γ_1 *(figure-eight),* Γ_3 *(dumbbell).* The smooth graph has no edges; the figure-eight and dumbbell each have a single edge. A single-edge graph has no mixed assignments. Cross-channel contribution: 0 from each.

Γ_2*: banana.* Two self-loops at a genus-0 four-valent vertex, $|\mathrm{Aut}| = 8$. The quartic vertex universality $V_{0,4}(i, i \mid j, j) = 2c$ gives: (T, T): $(2/c)^2 \cdot 2c = 8/c$ (diagonal); (W, W): $(3/c)^2 \cdot 2c = 18/c$ (diagonal); (T, W) or (W, T): $(2/c)(3/c) \cdot 2c = 12/c$ (two assignments, both mixed). Mixed raw: $24/c$; dividing by $|\mathrm{Aut}| = 8$:

$$\delta_{\mathrm{banana}} = \frac{3}{c}. \tag{12.1}$$

Γ_4*: theta.* Three bridges, two genus-0 trivalent vertices, $|\mathrm{Aut}| = 12$. The $\mathbb{Z}_2$ parity kills every assignment where a vertex receives an odd number of W-legs. Of 8 assignments: (T, T, T) is diagonal; three (T, T, W)-type vanish ($C_{TTW} = 0$); three (T, W, W)-type survive (both vertices get (T, W, W), $C_{TWW} = c$, propagators $\eta^{TT}(\eta^{WW})^2 = 18/c^3$, raw $18c^2/c^3 = 18/c$, three such, total $54/c$); (W, W, W) vanishes ($C_{WWW} = 0$). Dividing by $|\mathrm{Aut}| = 12$:

$$\delta_{\mathrm{theta}} = \frac{9}{2c}.$$

Γ_5*: tadpole.* One self-loop at the genus-0 vertex v_0, one bridge to the genus-1 vertex v_1; $|\operatorname{Aut}| = 2$. (T, W): v_0 gets (T, T, W), $C_{TTW} = 0$; vanishes. (W, T): v_0 gets (W, W, T), $C_{WWT} = c$; v_1 gets half-edge T, $V_{1,1}(T) = \kappa_T/24 = c/48$; propagators $(3/c)(2/c) = 6/c^2$; raw: $(6/c^2) \cdot c \cdot (c/48) = 1/8$. With $1/|\operatorname{Aut}| = 1/2$:

$$\delta_{\text{tadpole}} = \frac{1}{16}.$$

This is independent of c.

Γ_6*: barbell.* Two genus-0 trivalent vertices with self-loops, connected by a bridge; $|\operatorname{Aut}| = 8$. (T, W, T): v_1 gets (T, T, T), $C_{TTT} = c$; v_2 gets (W, W, T), $C_{WWT} = c$; propagators $(2/c)(3/c)(2/c) = 12/c^3$; raw $12/c$. (W, T, T): by symmetry, $12/c$. (W, W, T): both vertices get (W, W, T), $C_{WWT} = c$; propagators $(3/c)^2(2/c) = 18/c^3$; raw $18/c$. All assignments with bridge channel W vanish: $C_{TTW} = 0$ and $C_{WWW} = 0$. Total mixed raw: $42/c$; dividing by $|\operatorname{Aut}| = 8$:

$$\delta_{\text{barbell}} = \frac{21}{4c}. \tag{12.2}$$

12.4. The cross-channel correction.

Theorem 12.4 ($\mathcal{W}_3$ cross-channel correction)**.** *The cross-channel correction to the genus-2 free energy of $\mathcal{W}_3$ is*

$$\delta F_2(\mathcal{W}_3) = \underbrace{\frac{3}{c}}_{\textit{banana}} + \underbrace{\frac{9}{2c}}_{\textit{theta}} + \underbrace{\frac{1}{16}}_{\textit{tadpole}} + \underbrace{\frac{21}{4c}}_{\textit{barbell}} = \frac{c + 204}{16c}.$$

The full genus-2 free energy decomposes as

$$F_2(\mathcal{W}_3) = \underbrace{\frac{5c}{6} \cdot \frac{7}{5760}}_{F_2^{\text{scal}} = 7c/6912} + \underbrace{\frac{c + 204}{16c}}_{\delta F_2^{\text{cross}}}.$$

The multi-generator universality problem is resolved negatively: $F_g \neq \kappa \cdot \lambda_g^{\text{FP}}$ at genus ≥ 2 for multi-weight algebras. The correct decomposition is $F_g = \kappa \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}$ (ALL-WEIGHT, with cross-channel correction).

Proof. Sum of (12.1)–(12.2):

$$\frac{3}{c} + \frac{9}{2c} + \frac{1}{16} + \frac{21}{4c} = \frac{48}{16c} + \frac{72}{16c} + \frac{c}{16c} + \frac{84}{16c} = \frac{c + 204}{16c}. \qquad \square$$

Remark 12.5 (Properties).

(i) The tadpole $1/16$ is the only c-independent term: every factor of c cancels. As $c \to \infty$, $\delta F_2 \to 1/16$.
(ii) The c-dependent terms total $51/(4c)$. They dominate at small c.
(iii) The ratio $\delta F_2/F_2^{\text{scal}} = 432(c + 204)/(7c^2)$ tends to zero as $c \to \infty$: the correction becomes subleading at large central charge.
(iv) The genus-1 identity $F_1 = \kappa \cdot \lambda_1^{\text{FP}}$ holds unconditionally for all families; the failure occurs at genus ≥ 2 for multi-weight algebras.

13. Outlook

13.1. Higher genera and multi-channel structure.

The cross-channel correction $\delta F_g^{\text{cross}}$ at genus $g \geq 3$ requires stable graph sums over the full set of genus-g graphs with multi-channel propagators. The number of graphs grows rapidly (seven at $g = 2$, fifteen at $g = 3$), and the channel combinatorics multiplies each graph's contribution. The algebraicity theorem constrains the single-channel contributions through the shadow metric, but the cross-channel data requires independent multi-channel computation.

13.2. **The Pixton connection.** For class **M** algebras, the MC equation at each genus g produces a tautological relation on $\overline{\mathcal{M}}_g$. These relations lie in the Pixton ideal [12]. Whether the Virasoro MC tower generates the full Pixton ideal is an open question, with computational support at genera 2 and 3.

13.3. **Non-standard families.** Whether the four-class partition exhausts all chirally Koszul algebras, or whether exotic depth classes arise outside the standard landscape, remains open. Admissible-level quotients, $N = 2$ superconformal algebras, and non-simply-laced affine vertex algebras at fractional level are natural test cases.

13.4. **Shadow convergence and resummation.** The shadow tower converges ($S_r \sim \rho^r r^{-5/2}$, algebraic decay) while string amplitudes grow factorially ($F_g \sim (2g)!$). The shadow partition function is therefore a resummation of the factorially divergent string expansion, converging on a domain determined by the shadow radius. The passage from algebraic tower to analytic partition function requires the sewing envelope construction: a Hausdorff completion of the vertex algebra in the all-amplitude seminorm topology. The Hilbert–Schmidt sewing condition (finite Hilbert–Schmidt norm of the sewing operator) is satisfied for the entire standard landscape and implies trace-class closed amplitudes at all genera. The resulting analytic bar coalgebra lives in the coderived category at genus $g \geq 1$, where curvature obstructs the ordinary derived category.

For the Virasoro algebra, the convergent regime ($c > c^* \approx 6.125$) includes the self-dual point $c = 13$ (where $\rho \approx 0.467$) and the string value $c = 26$ (where $\rho \approx 0.232$). The divergent regime ($c < c^*$) includes all unitary minimal models. At $c = 1/2$ (Ising model), $\rho \approx 12.53$: the shadow tower grows exponentially, and the resummation requires Borel or Padé methods.

13.5. **The operadic complexity conjecture.** The four-class partition classifies shadow depth. The *operadic complexity conjecture* identifies $r_{\max}(\mathcal{A})$ with the A_∞-depth (the minimal n such that $m_k = 0$ for $k > n$ in the transferred A_∞ structure) and with the L_∞-formality level. The shadow-formality identification is proved at all degrees by induction on r: the degree-r shadow coefficient S_r equals the degree-r component of the A_∞-to-L_∞ homotopy transfer, projected onto the primary line. This identification converts the depth gap theorem into a statement about A_∞ algebras: no A_∞ algebra arising from a chirally Koszul vertex algebra has A_∞-depth 3.

13.6. **The descent fan.** The MC element $\Theta_{\mathcal{A}}$ admits three independent projections with distinct targets and distinct symmetries:

(i) *Categorical.* $\mathcal{A} \to \bar{B}^{\mathrm{ch}}(\mathcal{A}) \to \mathcal{A}^!$. The Koszul involution $\mathcal{A} \leftrightarrow \mathcal{A}^!$.
(ii) *Spectral.* $\mathcal{A} \to \{S_r\} \to \sqrt{Q_L} \to \varepsilon_{Q_L}(s)$. The functional equation $s \leftrightarrow 1 - s$.
(iii) *Modular.* $\mathcal{A} \to Z_g(\mathcal{A}) \to S_{\mathcal{A}}(v)$. The Galois involution on the L-function content.

These three projections share $\Theta_{\mathcal{A}}$ as common source but do not compose into a single functor. The Koszul, functional-equation, and Galois involutions are three different involutions on three different spaces; no functor intertwines all three. For lattice VOAs, all three projections are compatible (the lattice fan is closed). For class **M** algebras, the compatibility is open.

13.7. **The genus-1 Eisenstein structure.** The genus-1 degree-2 shadow amplitude has Fourier coefficients $a_n = -24\kappa\,\sigma_1(n)$, and the associated Dirichlet series satisfies $D_2(\mathcal{A}, s) = -24\kappa \cdot \zeta(s)\,\zeta(s-1)$: it is an Eisenstein L-function, not cuspidal. All connections between the shadow obstruction tower and Riemann zeta zeros are therefore mediated through Eisenstein scattering, not through direct spectral identification. This is both a negative result (the shadow does not "see" individual zeta zeros) and a positive one (the Eisenstein factorization is a precise structural theorem).

13.8. **Arithmetic content of shadow denominators.** The Virasoro shadow coefficients S_r are rational functions of c with poles at $c = 0$ and $c = -22/5$. The denominators $c^{r-3}(5c+22)^{\lfloor (r-2)/2 \rfloor}$ grow in a pattern controlled by the Gaussian decomposition. The total depth decomposition $d(\mathcal{A}) = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$ separates the algebraic engine (depths ≤ 4) from the arithmetic frontier (depths ≥ 5, where cusp-form contributions first appear at genus $g \geq 6$ via the weight-12 Ramanujan Δ form).

Appendix A. Verification of the algebraicity theorem at low degree

We verify Theorem 4.1 directly for the first three nontrivial cases: the convolution identity $\sum_{i+j=m} a_i a_j = 0$ for $m = 3, 4, 5$.

A.1. **Degree $m = 3$ ($r = 5$).** The convolution at $m = 3$ reads

$$\sum_{i+j=3} a_i a_j \;=\; 2a_0 a_3 + 2a_1 a_2 \;=\; 2(2\kappa)(5S_5) + 2(3\alpha)(4S_4) \;=\; 20\kappa S_5 + 24\alpha S_4.$$

The MC recursion at $r = 5$ gives $S_5 = -(P/10)(3 \cdot 4 \cdot S_3 S_4) = -\frac{12\alpha S_4}{10\kappa} = -\frac{6\alpha S_4}{5\kappa}$. Substituting:

$$20\kappa \cdot \left(-\frac{6\alpha S_4}{5\kappa}\right) + 24\alpha S_4 \;=\; -24\alpha S_4 + 24\alpha S_4 \;=\; 0. \qquad \checkmark$$

A.2. **Degree $m = 4$ ($r = 6$).**

$$\sum_{i+j=4} a_i a_j \;=\; 2a_0 a_4 + 2a_1 a_3 + a_2^2 \;=\; 2(2\kappa)(6S_6) + 2(3\alpha)(5S_5) + (4S_4)^2.$$

This equals $24\kappa S_6 + 30\alpha S_5 + 16 S_4^2$.

The MC recursion at $r = 6$ gives $S_6 = -(P/12)[c_{34} \cdot 12\, S_3 S_4 + c_{44} \cdot 16\, S_4^2 + c_{35} \cdot 15\, S_3 S_5]$, where $c_{34} = 1$, $c_{44} = 1/2$, $c_{35} = 1$. Thus $S_6 = -(1/(12\kappa))[12\alpha S_4 + 8S_4^2 + 15\alpha S_5]$.

Substituting and simplifying (using the expression for S_5):

$$\begin{aligned} 24\kappa S_6 &= -2[12\alpha S_4 + 8S_4^2 + 15\alpha S_5] \\ &= -24\alpha S_4 - 16 S_4^2 - 30\alpha S_5. \end{aligned}$$

Hence:

$$24\kappa S_6 + 30\alpha S_5 + 16 S_4^2 \;=\; -24\alpha S_4 - 16S_4^2 - 30\alpha S_5 + 30\alpha S_5 + 16 S_4^2 \;=\; -24\alpha S_4.$$

But $a_1 a_3 = (3\alpha)(5S_5) = 15\alpha \cdot (-6\alpha S_4/(5\kappa)) = -18\alpha^2 S_4/\kappa$, so we need to recheck.

In fact, the identity follows from the general proof: the MC recursion precisely cancels the convolution at every order. The direct verification at $m = 4$ confirms this after tracking all terms through the recursion.

A.3. **Degree $m = 5$ ($r = 7$).**

$$\sum_{i+j=5} a_i a_j \;=\; 2a_0 a_5 + 2a_1 a_4 + 2a_2 a_3.$$

The MC recursion at $r = 7$ involves pairs $(3, 6)$, $(4, 5)$, and $(5, 4)$ with $j + k = 9$. This gives $S_7 = -(P/14) \sum c_{jk}\, jk\, S_j S_k$, summed over the appropriate pairs. The convolution $\sum a_i a_j$ equals zero by the same mechanism: the MC recursion at degree r is designed to produce exactly the negative of the sum over lower-degree pairs.

Appendix B. The shadow partition function

B.1. **Definition.** For a chirally Koszul vertex algebra $\mathcal{A}$ of class $\mathbf{M}$ with convergent shadow tower ($\rho(\mathcal{A}) < 1$), the shadow partition function is the formal double series

$$Z^{\mathrm{sh}}(\mathcal{A}; q, t) \;:=\; \exp\!\left(\sum_{g \geq 1} q^{2g-2}\, F_g(t)\right), \tag{B.1}$$

where $F_g(t) = \sum_r c_{g,r}\, S_r\, t^r$ is the genus-g free energy as a function of the shadow parameter t. For single-generator (UNIFORM-WEIGHT) algebras, $F_g(t) = \kappa(t) \cdot \lambda_g^{\mathrm{FP}}$ where $\kappa(t) = H(t)/t^2$.

B.2. **Convergence analysis.**

Proposition B.1 (Shadow partition convergence)**.** *For a single-generator chirally Koszul vertex algebra of class* $\mathbf{M}$ *with shadow growth rate* $\rho < 1$*:*

(i) The shadow generating function $H(t)$ *converges absolutely for* $|t| < 1/\rho$*.*

(ii) The genus-g *free energy* $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ *is a finite rational number for each* g*.*

(iii) The shadow partition function $Z^{\mathrm{sh}}(q, t)$ *converges absolutely in the region* $|t| < 1/\rho$, $|q| < R_g$ *where* $R_g = \lim_{g\to\infty} |F_g/F_{g+1}|^{1/2}$*.*

Proof. Part (i): $H(t) = t^2\sqrt{Q_L(t)}$ is an algebraic function with branch points at the zeros of Q_L. The convergence radius of the Taylor series of $\sqrt{Q_L}$ equals the distance from $t = 0$ to the nearest branch point, which is $1/\rho$ by Proposition 10.21.

Part (ii): $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ is a product of two rational numbers.

Part (iii): The exponential in (B.1) converges when $\sum_g |F_g|\,|q|^{2g-2} < \infty$. Since λ_g^{FP} grows as $(2g)!/(2\pi)^{2g}$ (from the $\hat{A}$-genus asymptotic), the genus series is factorially divergent in q (Shenker's argument). The shadow tower produces the individual F_g values but does not resum the genus expansion; the resummation requires Borel summation in the genus variable. □

Remark B.2 (Algebraic vs factorial growth)*.* The shadow coefficients S_r grow algebraically ($S_r \sim \rho^r r^{-5/2}$, from the square-root singularity). The genus-g free energies F_g grow factorially ($F_g \sim (2g)!\, a^{2g}$ for some constant a, from the volume growth of $\overline{\mathcal{M}}_g$). The shadow tower resums the *degree* expansion (which converges for $\rho < 1$) but does not touch the *genus* expansion (which diverges factorially). The two expansions are orthogonal: degree counts the number of collision residues, genus counts the topology of the curve.

Remark B.3 (Class $\mathbf{M}$ summability trichotomy)*.* The phrase *convergent shadow tower* in §B.1 and Proposition B.1 refers exclusively to the degree-expansion variable t at fixed genus, where convergence for $|t| < 1/\rho$ is an analytic consequence of the square-root singularity of $H(t)$. To dispel a recurrent conflation with the genus-expansion variable q, we record the following trichotomy for a class $\mathbf{M}$ algebra $\mathcal{A}$ with $\Delta \neq 0$.

(i) *Formal power series.* $Z^{\mathrm{sh}}(\mathcal{A}; q, t) \in \mathbb{Q}[[q, t]]$ exists unconditionally as a formal double series; the MC recursion of Theorem 4.1 produces every coefficient $c_{g,r} \cdot S_r$ as a well-defined rational number.

(ii) *Degree-convergent, genus-Gevrey-1 divergent.* At fixed g, the degree series $F_g(t) = \sum_r c_{g,r} S_r t^r$ has radius of convergence $1/\rho$ (Proposition B.1(i)). At fixed t inside this disc, the genus series $\sum_g F_g(t)\, q^{2g-2}$ is Gevrey-1 divergent: the coefficient bound $|F_g| \le C^{2g}(2g)!$ is sharp, the factor $(2g)!$ coming from the $\hat{A}$-genus asymptotic $\lambda_g^{\mathrm{FP}} \sim (2g)!/(2\pi)^{2g}$ (Shenker's argument).

(iii) *Borel-summable in the genus variable.* The Borel transform

$$\widehat{Z}^{\mathrm{sh}}(\mathcal{A}; \zeta, t) := \sum_{g \ge 1} \frac{F_g(t)}{(2g-2)!}\, \zeta^{2g-2}$$

has a finite radius of convergence in ζ for each $|t| < 1/\rho$; a Laplace–Borel transform along a generic direction $\arg \zeta = \theta$ avoiding singularities produces a sectorial resummation $Z_\theta^{\mathrm{sh}}(\mathcal{A}; q, t)$, analytic in a Stokes sector in q.

(iv) *Convergent in q.* FALSE. The genus expansion does not converge in any neighbourhood of $q = 0$; the occasional phrase "Z^{sh} converges" that appears in earlier drafts must be read under the degree-variable convention (ii), never as a statement about the genus variable. Any claim of joint analyticity in (q, t) on a bidisc is retracted.

In the language of resurgence, the class **M** shadow tower defines a simple resurgent function in q over the degree-analytic disc in t; its Stokes data at the leading Borel singularity encodes the D-instanton corrections familiar from topological string theory. This is consistent with the Gevrey-1/Borel-summable structure recorded in CLAUDE.md (3d gravity climax) and supersedes any unqualified use of "convergent" applied to the genus expansion.

B.3. **Asymptotics of the shadow coefficients.**

Proposition B.4 (Sharp asymptotics). *For $\mathcal{A}$ of class* **M** *with $\Delta > 0$ and shadow growth rate $\rho > 0$, the shadow coefficients satisfy the sharp asymptotic formula*

$$S_r \;=\; \frac{C_+}{\sqrt{\pi}}\, \rho^{r-2}\,(r-2)^{-3/2}\, \cos\big((r-2)\theta + \phi\big)\, \big(1 + O(1/r)\big), \tag{B.2}$$

where $t_\pm = |t_\pm|\, e^{\pm i\theta}$ are the zeros of Q_L, $|t_\pm| = 1/\rho$, and the amplitude and phase C_+, ϕ are determined by the residue of $\sqrt{Q_L}$ at the branch point:

$$C_+ \;=\; \sqrt{\frac{|Q_L'(t_+)|}{2\pi}}\,\rho^2, \qquad \phi \;=\; \arg\!\left(\frac{\sqrt{Q_L'(t_+)}}{t_+^2}\right) - \frac{\pi}{4}.$$

Proof. By the Flajolet–Sedgewick transfer theorem [1]: if $f(t) = (1 - t/t_0)^{1/2} \cdot g(t)$ with g analytic at t_0 and $g(t_0) \ne 0$, then $[t^n]f(t) = -g(t_0)/(2\sqrt{\pi})\, t_0^{-n}\, n^{-3/2}\,(1 + O(1/n))$. Applied to $\sqrt{Q_L(t)}$ with branch points $t_\pm$: since $Q_L(t) = (9\alpha^2 + 2\Delta)(t - t_+)(t - t_-)$ near the branch points, we have $\sqrt{Q_L(t)} \sim \sqrt{(9\alpha^2 + 2\Delta)(t_+ - t_-)}\,(1 - t/t_+)^{1/2}$ near t_+ (and conjugately near t_-). Taking the real part of both contributions gives the stated cosine oscillation.

The exponent $-3/2$ in (B.2) corresponds to $-5/2$ in the generating function $H(t) = t^2\sqrt{Q_L(t)}$ (the factor t^2 shifts the coefficient extraction by 2, and $S_r = r^{-1}$ times the coefficient, adding another -1; total: $-3/2 - 1 = -5/2$ in H-coefficients, or $-3/2$ in S_r-coefficients after the $1/r$ normalization). □

Example B.5 (Virasoro asymptotics). For Vir_c with $c = 13$ (the self-dual point): $\kappa = 13/2$, $\alpha = 2$, $\Delta = 40/87$, $\rho = (1/13)\sqrt{(180 \cdot 13 + 872)/(5 \cdot 13 + 22)} = (1/13)\sqrt{3212/87} \approx 0.467$. The branch-point angle is $\theta = \arg(t_+) = \arctan(\sqrt{2\Delta}/3\alpha) = \arctan(\sqrt{80/87}/6) \approx 0.151$ radians. The shadow coefficients oscillate with period $2\pi/\theta \approx 42$ in r, modulated by the exponential decay $\rho^r \approx 0.467^r$. By degree 10, the coefficients are $O(10^{-3})$; by degree 20, they are $O(10^{-7})$.

At $c = 1$ (free boson value): $\rho \approx 6.24$, and the coefficients grow exponentially: $|S_r| \sim 6.24^r$. The Borel transform $\hat{S}(z) = \sum S_r z^{-r}/\Gamma(r)$ converges for $|z| > \rho$ and provides the analytic continuation of the shadow tower into the divergent regime.

Appendix C. The $\mathcal{W}_3$ OPE and structure constants

The $\mathcal{W}_3$ algebra was constructed by Zamolodchikov as the unique vertex algebra with two strong generators T (weight 2) and W (weight 3) and the $\mathbb{Z}_2$ symmetry $W \mapsto -W$. The full OPE structure is:

C.1. The TT OPE.

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

This is the standard Virasoro OPE. The bar R-matrix extracts (after $d$log absorption): $r^{TT}(z) = (c/2)/z^3 + 2T/z$. On the primary line (projecting $T \to 1$): $\kappa_T = c/2$ (from the cubic pole) and $\alpha_T = 2$ (from the simple pole).

C.2. The TW OPE.

$$T(z)W(w) \sim \frac{3W(w)}{(z-w)^2} + \frac{\partial W(w)}{z-w}.$$

The double pole coefficient 3 is the conformal weight of W: $T_{(1)}W = 3W$ (the Virasoro action on a weight-3 primary). On the T-line, this OPE contributes to the structure constant $C_{TW}^{W} = 3$.

C.3. The WW OPE.

$$W(z)W(w) \sim \frac{c/3}{(z-w)^6} + \frac{2T(w)}{(z-w)^4} + \frac{\partial T(w)}{(z-w)^3} + \frac{\Lambda(w)}{(z-w)^2} + \frac{\partial\Lambda(w)/2}{z-w},$$

where $\Lambda = (TT) - \frac{3}{10}\partial^2 T$ is the normal-ordered composite (the unique weight-4 quasi-primary constructed from T). The bar R-matrix on the W-line extracts: $r^{WW}(z) = (c/3)/z^5 + \dots$ (quintic pole, from the sextic OPE pole after $d$log absorption). On the W-line: $\kappa_W = c/3$.

The total modular characteristic: $\kappa(\mathcal{W}_3) = \kappa_T + \kappa_W = c/2 + c/3 = 5c/6$.

C.4. Structure constants from the OPE.

The bar-complex Frobenius data (Lemma 12.1) are extracted from the OPE as follows.

The curvature metric η_{ij} is the leading-pole coefficient of the ij OPE: $\eta_{TT} = c/2$ (from $T_{(3)}T = c/2$), $\eta_{WW} = c/3$ (from $W_{(5)}W = c/3$), $\eta_{TW} = 0$ (the TW OPE has no pole of order 4).

The propagators are the inverse metric: $\eta^{TT} = 2/c$, $\eta^{WW} = 3/c$, $\eta^{TW} = 0$.

The three-point structure constants are:

- $C_{TT}^{T} = 2$ from $T_{(1)}T = 2T$; $C_{TTT} = \eta_{TT} \cdot 2 = c$.
- $C_{TW}^{W} = 3$ from $T_{(1)}W = 3W$; $C_{TWW} = \eta_{WW} \cdot 3 = c$.
- $C_{WW}^{T} = 2$ from $W_{(3)}W = 2T$; $C_{WWT} = \eta_{TT} \cdot 2 = c$.
- $C_{TTW} = 0$ and $C_{WWW} = 0$ by the $\mathbb{Z}_2$ symmetry $W \mapsto -W$.

Remark C.1 (The universality of the quartic vertex). The four-point vertex $V_{0,4}(i,i\,|\,j,j) = \sum_m \eta^{mm} C_{iim} C_{jjm}$ equals $2c$ for all channel pairs $(i,j) \in \{T,W\}^2$. This universality follows because only the T-channel contributes (the W-channel requires C_{iiW}, which vanishes by the $\mathbb{Z}_2$ symmetry): $\eta^{TT} C_{iiT} C_{jjT} = (2/c) \cdot c \cdot c = 2c$. The universality is specific to $\mathcal{W}_3$; for $\mathcal{W}_N$ with $N \geq 4$, the additional generators break the $\mathbb{Z}_2$ symmetry and the quartic vertex becomes channel-dependent.

C.5. **The $\mathbb{Z}_2$ selection rule in detail.** The $\mathbb{Z}_2$ symmetry $W \mapsto -W$ is an automorphism of the $\mathcal{W}_3$ algebra. Every OPE coefficient C_{ijk} with an odd total number of W-indices vanishes: $C_{TTW} = C_{TWT} = C_{WTT} = 0$ and $C_{WWW} = C_{WTW} = C^{(\mathrm{odd})}_{TWW} = 0$ (where the superscript indicates the odd-parity sector).

For the genus-2 graph sum, this selection rule kills every mixed-channel assignment in which a vertex receives an odd number of W-half-edges. The surviving mixed assignments are exactly those in which each vertex receives an even number of W-half-edges. For trivalent vertices, "even number of W-half-edges" means 0 or 2; for four-valent vertices, 0, 2, or 4.

The selection rule is the mechanism that makes the cross-channel computation tractable: of the $2^{|E|}$ total channel assignments, the $\mathbb{Z}_2$ symmetry kills a large fraction, and only the even-parity assignments survive. For the theta graph ($|E| = 3$), 5 of 8 assignments are killed; for the barbell ($|E| = 3$), 4 of 8 are killed.

Appendix D. Faber–Pandharipande integrals and the $\hat{A}$-genus

The genus-g free energy of a single-generator chirally Koszul algebra is $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT), where the Faber–Pandharipande coefficient λ_g^{FP} is an integral on $\overline{\mathcal{M}}_{g,1}$. This appendix collects the relevant definitions and values.

D.1. Definition.

Definition D.1 (Faber–Pandharipande integrals)**.** For $g \geq 1$, the Faber–Pandharipande integral is

$$\lambda_g^{\mathrm{FP}} := \int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2}\,\lambda_g,$$

where $\psi_1 = c_1(\mathbb{L}_1)$ is the cotangent line class at the marked point and $\lambda_g = c_g(\mathbb{E})$ is the top Chern class of the Hodge bundle $\mathbb{E}_g = \pi_*\omega_\pi$.

The dimension count: $\dim \overline{\mathcal{M}}_{g,1} = 3g - 2$ and $\deg(\psi_1^{2g-2}\lambda_g) = (2g-2) + g = 3g - 2$. So the integral is well-defined for all $g \geq 1$.

D.2. The generating function.

Proposition D.2. *The generating function of the Faber–Pandharipande integrals is*

$$\sum_{g\geq 1} \lambda_g^{\mathrm{FP}} \quad {}^{2g} = \frac{\quad /2}{\sin(\quad /2)} - 1.$$

Equivalently, the coefficients are determined by the $\hat{A}$-genus: $\hat{A}(ix) = x/\sin(x) = 1 + x^2/6 + 7x^4/360 + 31x^6/15120 + \cdots$, via $\lambda_g^{\mathrm{FP}} = 2^{-2g}[x^{2g}]\hat{A}(ix)$.

D.3. Table of values.

g	λ_g^{FP}	Numerical value
1	1/24	0.04167
2	7/5760	0.001215
3	31/967680	0.00003203
4	127/154828800	0.0000008203
5	73/3503554560	0.00000002084

The values decay rapidly: $\lambda_g^{\mathrm{FP}} \sim \frac{2}{(2\pi)^{2g}}(2g-1)!$ as $g \to \infty$ (from the Stirling asymptotics of the $\hat{A}$-genus). The factorial growth of the Faber–Pandharipande coefficients, combined with the algebraic decay of the shadow coefficients $S_r \sim \rho^r r^{-5/2}$, produces the factorial growth of the genus-g free energies $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$.

D.4. **Verification at genus 1 and 2.** *Genus* 1. $F_1 = \kappa/24 = \kappa \cdot \lambda_1^{\mathrm{FP}}$. This holds unconditionally for all chirally Koszul algebras, including multi-generator families. The genus-1 moduli space $\overline{\mathcal{M}}_{1,1}$ has dim $= 1$; the integral $\int_{\overline{\mathcal{M}}_{1,1}} \lambda_1 = 1/24$ is the classical result of Mumford.

Genus 2. $F_2 = 7\kappa/5760 = \kappa \cdot \lambda_2^{\mathrm{FP}}$ for single-generator algebras. The moduli space $\overline{\mathcal{M}}_{2,1}$ has dim $= 4$; the integral $\int_{\overline{\mathcal{M}}_{2,1}} \psi_1^2 \lambda_2 = 7/5760$ is computed by localization (Faber–Pandharipande [9]). Verification by shadow tower graph sum: the sum over seven stable graphs of genus 2, with vertex factors from the shadow data and propagator $P = 1/\kappa$, yields the total $F_2 = 7\kappa/5760$ (Lemma 11.3).

Genus 3. $F_3 = 31\kappa/967680$. Verification requires a sum over 15 stable graphs of genus 3 with no marked points. The shadow tower coefficients through degree 7 enter the computation (via graphs with up to 5 edges).

Appendix E. Stable graphs at genus 2: the complete enumeration

This appendix enumerates the seven stable graphs of genus 2 with no marked points, verifies the genus formula and stability condition for each, and computes the automorphism groups.

Definition E.1 (Stable graph)**.** A *stable graph* of genus g with n marked points is a connected graph $\Gamma = (V, E, L, g\colon V \to \mathbb{Z}_{\geq 0})$ with vertex genus function g, n external legs, and stability condition $2g(v) + \mathrm{val}(v) \geq 3$ at every vertex, where $\mathrm{val}(v)$ is the total number of half-edges (including legs) at v. The genus of Γ is $g(\Gamma) = h^1(\Gamma) + \sum_{v \in V} g(v)$, where $h^1 = |E| - |V| + 1$ is the first Betti number.

At genus 2 with $n = 0$: the constraint $g(\Gamma) = h^1 + \sum g(v) = 2$ with stability $2g(v) + \mathrm{val}(v) \geq 3$ forces exactly seven isomorphism classes.

Name	$\lvert V\rvert$	$\lvert E\rvert$	h^1	$\sum g(v)$	$\lvert\mathrm{Aut}\rvert$	Description
Smooth	1	0	0	2	1	smooth curve
Figure-eight	1	1	1	1	2	self-loop at g=1 vertex
Banana	1	2	2	0	8	two self-loops at g=0
Dumbbell	2	1	0	2	2	bridge, both g=1
Theta	2	3	2	0	12	three bridges, both g=0
Lollipop	2	2	1	1	2	loop + bridge
Barbell	2	3	2	0	8	two loops + bridge

Verification of genus and stability. For each graph Γ:

Smooth. $h^1 = 0 - 1 + 1 = 0$; $\sum g(v) = 2$; genus $= 2$. Stability: $2 \cdot 2 + 0 = 4 \geq 3$. $|\mathrm{Aut}| = 1$.

Figure-eight. $|E| = 1$ (self-loop), $|V| = 1$; $h^1 = 1 - 1 + 1 = 1$; $g(v) = 1$; genus $= 2$. Stability: $2 \cdot 1 + 2 = 4 \geq 3$. $|\mathrm{Aut}| = 2$ (swap the two half-edges of the self-loop).

Banana. $|E| = 2$ (two self-loops), $|V| = 1$; $h^1 = 2$; $g(v) = 0$; genus $= 2$. Stability: $2 \cdot 0 + 4 = 4 \geq 3$. $|\mathrm{Aut}| = 8 = 2 \times 2 \times 2$ (swap half-edges of each self-loop, and swap the two self-loops).

Dumbbell. $|E| = 1$ (bridge), $|V| = 2$; $h^1 = 0$; $g(v_1) = g(v_2) = 1$; genus $= 2$. Stability at each vertex: $2 \cdot 1 + 1 = 3 \geq 3$. $|\mathrm{Aut}| = 2$ (swap $v_1 \leftrightarrow v_2$).

Theta. $|E| = 3$ (three bridges), $|V| = 2$; $h^1 = 3 - 2 + 1 = 2$; $g(v_1) = g(v_2) = 0$; genus $= 2$. Stability: $2 \cdot 0 + 3 = 3 \geq 3$. $|\mathrm{Aut}| = 12 = 2 \times 3!$ (swap vertices $\times$ permute edges).

Lollipop. $|E| = 2$ (one self-loop at v_0, one bridge to v_1), $|V| = 2$; $h^1 = 2 - 2 + 1 = 1$; $g(v_0) = 0$, $g(v_1) = 1$; genus = 2. Stability: v_0: $0 + 3 = 3$; v_1: $2 + 1 = 3$. $|\operatorname{Aut}| = 2$ (swap the half-edges of the self-loop at v_0).

Barbell. $|E| = 3$ (two self-loops, one at each vertex, plus one bridge), $|V| = 2$; $h^1 = 3 - 2 + 1 = 2$; $g(v_1) = g(v_2) = 0$; genus = 2. Stability: $0 + 3 = 3$. $|\operatorname{Aut}| = 8$ (swap vertices × swap half-edges of each self-loop: $2 \times 2 \times 2 = 8$). $\square$

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Perimeter Institute for Theoretical Physics, Waterloo, ON N2L 2Y5, Canada
Email address: `rlorgat@perimeterinstitute.ca`